



EPFL RACING TEAM
POWER ELECTRONICS DIVISION

PROFESSOR : DENIS GILLET // TEAM LEADER : ANTONIN STAMPBACH

Semester project report

STATE ESTIMATION AND TRACTION CONTROL FOR A FORMULA STUDENT
RACING CAR

VRAY Alexandre 310104

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Symbols

Annotations

q	Any variable
q_x	Longitudinal in car frame
q_y	Lateral in car frame
q_l	Longitudinal in wheel frame
q_{lat}	Lateral in wheel frame
q_i	Relative to the wheel with index i (FL,FR,RL,RR)
q_m	Motor variable
q_w	Wheel variable
$\tilde{q}$	Measure
$\hat{q}$	Estimate
$q(k)$	Discrete sampling at instant k
q^-/q^+	Before/After update

Variables

v	Speed
a	Acceleration
$\dot{\psi}$	Yaw rate
γ	wheel drive steer
δ	wheel steer
i	Current RMS
P	Pressure
τ	Torque
ω	Angular speed
F	Force
μ	Friction coefficient
s	Slip ratio
α	Slip Angle

Car parameters

Chassis

a	Front axle length
b	Rear axle length
l_f	Distance between the front axle and the center of gravity
l_r	Distance between the rear axle and the center of gravity
l	Distance between front and rear axles
m	Total mass (vehicle + body)
z_{CG}	Height of the center of gravity
I_{zz}	Inertia of the car around z axis

Aerodynamics

A_{front}	Front area
c_{drag}	Drag coefficient
c_{lift}	Lift coefficient
ρ_{air}	Air density

Motor

τ_{max}	Maximum applicable torque
η	Gear ratio

Wheel

I_w	Inertia
R_w	Radius
k_b	Brake friction coefficient
k_d	Dynamic friction coefficient
k_s	Static friction coefficient

Tyre

$BCD_s/C_s/\dots$	Longitudinal Pacejka coefficients
$BCD_\alpha/C_\alpha/\dots$	Lateral Pacejka coefficients

Estimation parameters

ξ_q	Uncertainty of a model equation
η_q	Uncertainty of a sensor measurement
σ_q	Standard deviation of these uncertainties

Traction control parameters

K_p	Proportional gain
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Control fusion parameters

w_i	Quadratic weight
$\mathbf{Q}$	Weight matrix
Σ_τ	Sum of the torques
Δ_τ	Difference of torques between left and right sides
Δ_τ^{ij}	Difference of torques between the wheel i and j respectively

1 Introduction

This semester project, conducted within the EPFL racing team [1], focuses on state estimation and traction control. However, before delving into these aspects, it is crucial to establish a comprehensive understanding of the vehicle dynamics and its inherent nonlinearities. Only with this understanding can efficient state estimation be performed, utilizing the multiple sensors available on the car and leveraging knowledge of the vehicle dynamics.

Accurate estimation of car velocity and wheel slips is essential for effective control strategies. Based on these estimations, the traction control system aims to track an optimal slip ratio, maximizing friction and preventing situations of excessive slip.

Furthermore, the project will explore the fusion of the traction control algorithm with torque vectoring and power limitation. This integration aims to optimize the overall control system by effectively coordinating traction control, torque vectoring, and power limiting functionalities.

It is important to note that this project builds upon the previous research conducted by Nathan Decurnex, Antonin Stampbach, and Willian Ducommun [2], [3]. Their work forms the foundation for this project and provides valuable insights into the subject matter.

2 Vehicle dynamics

The aim of this section is to acquire an effective nonlinear model of the racing car for various purposes. Firstly, having a simulation for control tests proves beneficial when the car is unavailable. Secondly, and possibly the most significant, it can validate the model utilized in subsequent stages of this project, ensuring that the identified parameters are accurate with a known level of uncertainty. Each aspect of the vehicle modeling is elaborated upon in the respective subsections provided below.

2.1 General schema

To ensure a clear understanding of each component in the vehicle modeling, it is important to appropriately name all variables without any ambiguity. Throughout this document, the convention used follows the representation shown in the following figure (Figure 1). Accordingly, the x axis aligns with the vehicle's longitudinal direction and is positive in the forward direction. The y axis corresponds to the lateral axis and is positive in the right direction. Lastly, the yaw rate $\dot{\psi}$ denotes a positive rotation along the z axis.

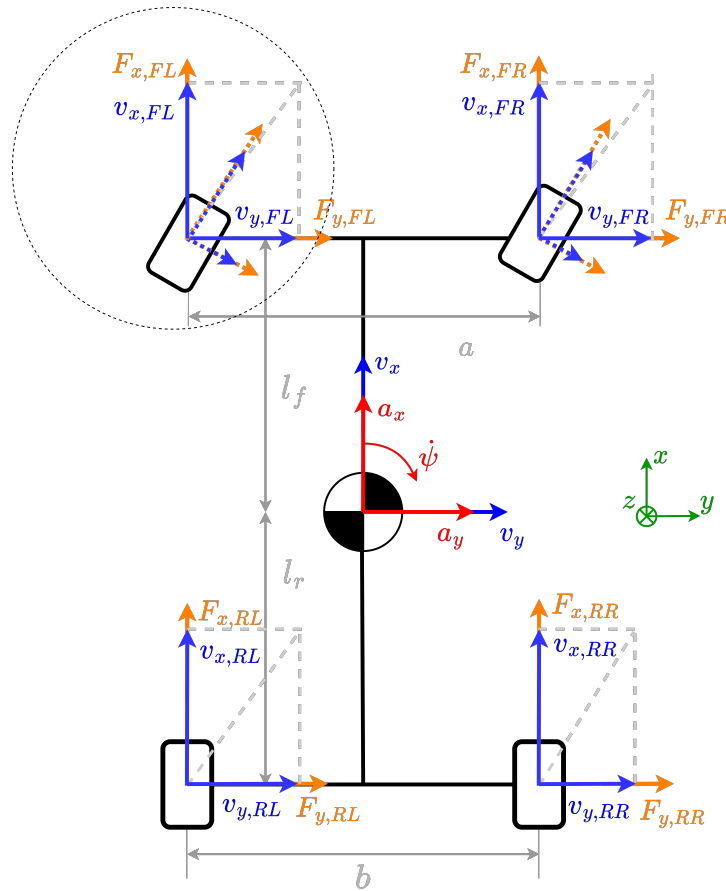


Figure 1: Simplified 4WD vehicle schema

To enhance understanding of the wheel variables, a closer examination is conducted, focusing on the wheel frame. In this context, the subscript x or y denotes the x and y axes of the body frames, while the subscripts l and lat are used to represent the longitudinal and lateral axes of the wheel frames, as illustrated in Figure 2.

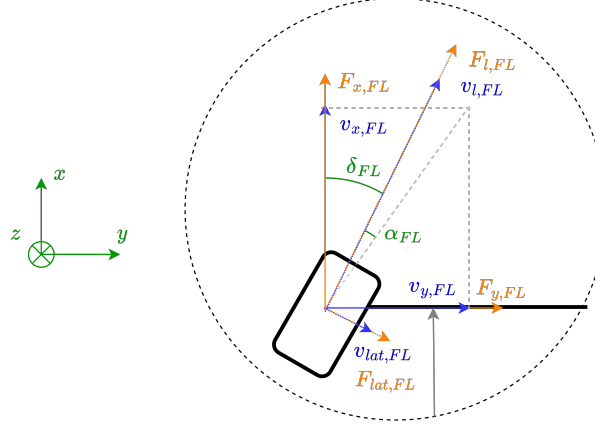


Figure 2: Simplified 4WD vehicle schema (wheel zoom)

2.2 General dynamics

The vehicle model begins by applying the second Newton's law and the angular momentum theorem. Due to the negligible lateral velocity, the lateral fluid friction is disregarded, and only the longitudinal fluid friction is taken into account. The resulting equations derived from these principles are presented in the following set of equations (1).

$$ma_x = \sum_{i=1}^4 F_{x,i} + F_{drag}, \quad ma_y = \sum_{i=1}^4 F_{y,i}, \quad I_{zz}\ddot{\psi} = \sum_{i=1}^4 (F_{y,i}p_{x,i} - F_{x,i}p_{y,i}). \quad (1)$$

Due to the steering of the wheels, the longitudinal and lateral forces in the body frame can be expressed in terms of the horizontal forces in the wheel frame, as shown in the following equation (2).

$$\begin{pmatrix} F_{x,i} \\ F_{y,i} \end{pmatrix} = \begin{pmatrix} \cos(\delta_i) & -\sin(\delta_i) \\ \sin(\delta_i) & \cos(\delta_i) \end{pmatrix} \cdot \begin{pmatrix} F_{l,i} \\ F_{lat,i} \end{pmatrix}. \quad (2)$$

Lastly, the positions of each wheel can be determined in terms of the car's geometric parameters, specifically the axle lengths illustrated in the general diagram (Figure 1). The positions can be derived using the equations (3):

$$\begin{cases} p_{x,FL} = l_f, & p_{y,FL} = \frac{a}{2} \\ p_{x,FR} = l_f, & p_{y,FR} = -\frac{a}{2} \\ p_{x,RL} = -l_r, & p_{y,RL} = \frac{b}{2} \\ p_{x,RR} = -l_r, & p_{y,RR} = -\frac{b}{2} \end{cases}. \quad (3)$$

2.3 Steering

In this subsection, the objective is to determine the angle of each wheel (Figure 2) based on the angle of the drive wheel γ , which is defined as positive in the clockwise direction (Figure 3).

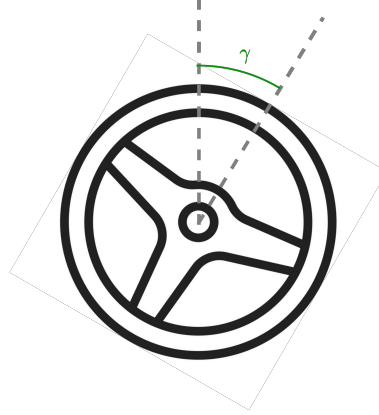


Figure 3: Driving wheel steer

Taking into account the geometry of the transmission, the wheel steer can be determined proportionally with the drive angle. If the angle is in the same direction as the wheel, an inner coefficient is applied; otherwise, an outer coefficient is used. This relationship is described in the equation (4).

$$\delta_{FL} = \begin{cases} k_{in}\gamma & , \text{if } \gamma < 0 \\ k_{out}\gamma & , \text{otherwise} \end{cases}, \quad \delta_{FR} = \begin{cases} k_{out}\gamma & , \text{if } \gamma < 0 \\ k_{in}\gamma & , \text{otherwise} \end{cases}. \quad (4)$$

2.4 Normal forces

In this section, a model for the normal forces is established, without taking into account the suspensions, as modeling them is considerably more challenging. Considering the significantly faster dynamics of the suspensions compared to the overall dynamics of the vehicle, this approximation is deemed sufficient for the general model. To calculate the normal forces, both the vehicle weight and the lift force are considered. However, the majority of variations in the normal forces stem from weight transfers. Thus, for each wheel, the normal force F_z is computed using the equation (5) provided below.

$$\begin{cases} F_{z,FL} = \frac{mgl_r}{2l} + \frac{ma_y z_{CG}}{2a} - \frac{ma_x z_{CG}}{2l} + F_{lift} \\ F_{z,FR} = \frac{mgl_r}{2l} - \frac{ma_y z_{CG}}{2a} - \frac{ma_x z_{CG}}{2l} + F_{lift} \\ F_{z,RL} = \frac{mgl_f}{2l} + \frac{ma_y z_{CG}}{2b} + \frac{ma_x z_{CG}}{2l} + F_{lift} \\ F_{z,RR} = \frac{mgl_f}{2l} - \frac{ma_y z_{CG}}{2b} + \frac{ma_x z_{CG}}{2l} + F_{lift} \end{cases}. \quad (5)$$

With,

$$F_{lift} = \frac{1}{4} \cdot \frac{1}{2} \cdot A \cdot C_{lift} \cdot \rho \cdot v_x^2, \quad a_x = \dot{v}_x - v_y \dot{\psi}, \quad a_y = \dot{v}_y + v_x \dot{\psi}.$$

2.5 Wheels

The wheel model needs to be precisely defined since experiments have demonstrated the significant impact that any parameter within this submodel can have on the final results. Initially, the idea was

to model the wheel dynamics by considering only the applied torques and the longitudinal force from the road (Figure 4). Consequently, the equation of dynamics was relatively straightforward and can be expressed as (6).

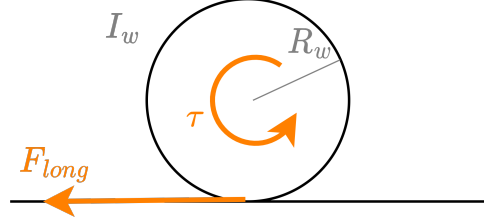


Figure 4: schema of the wheel dynamics

$$I_w \dot{\omega}_i = \tau_i - R_w F_{l,i} \quad (6)$$

The previous model's simulations did not produce satisfactory results. However, it was determined that the inclusion of the internal friction of the wheels is crucial in the model. This internal friction is described by a dynamic friction coefficient k_d and a static friction coefficient k_s . Furthermore, the model should also incorporate the effects of mechanical brake forces, where the counter brake torque is directly proportional to the applied brake pressure with a coefficient k_b . Consequently, the wheel dynamics, illustrated in Figure 5, can be expressed by the equations presented in the following form (7).

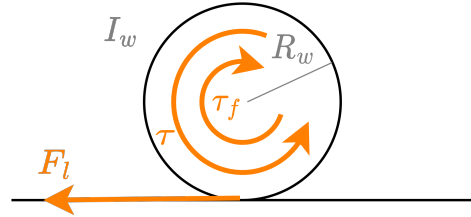


Figure 5: schema of the wheel dynamics

$$I_w \dot{\omega}_i = \begin{cases} 0 & \text{if } \omega = 0 \text{ and } |\tau_{f,i}| > |\tau_i - R_w F_{long,i}| \\ \tau_i - \tau_{f,i} - R_w F_{long,i} & \text{otherwise} \end{cases} \quad (7)$$

With,

$$\tau_{f,i} = k_b P_{b,i} + k_d \omega_i + k_s. \quad (8)$$

The specific values used for k_b , k_d , and k_s are provided in Table 1. To determine these values, one approach is to apply a torque to a wheel that is not in contact with the ground. Firstly, this is done without any brake pressure, and then repeated with brake pressure applied. By observing the resulting behavior, the values for k_b , k_d , and k_s can be identified.

k_b [m]	k_d [N.m.s]	k_s [Nm]
0.18	0.17	15

Table 1: internal wheel friction values

2.6 Slips

In this subsection, the slip ratio and slip angle of a tire are introduced. These terms will be frequently mentioned in subsequent sections and must be clearly defined. Slip refers to the difference between the angular velocity of the wheel multiplied by its radius and the longitudinal velocity. The slip ratio is then obtained by normalizing the slip value by the longitudinal speed. On the other hand, the slip angle represents the angle between the longitudinal direction of the wheel and the velocity vector. Figure 6 provides a visual representation of these slip variables, and the corresponding equations for their calculation are presented in (9).

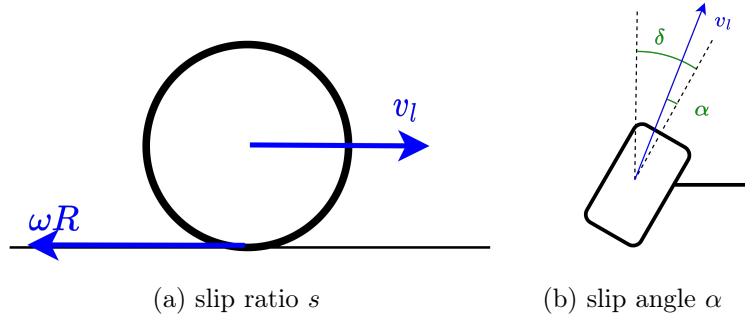


Figure 6: representations of the slip ratio and slip angle

$$s = \frac{\omega R - v_l}{v_l}, \quad \alpha = -\arctan\left(\frac{v_{lat}}{v_l}\right). \quad (9)$$

2.7 Tires

In the General Dynamics section (Section 2.2), the horizontal forces acting on the tires are assumed to be known. However, the question arises as to how these forces are computed. These forces are solely attributed to the friction between the tires and the road, and therefore can be calculated as the product of the normal force F_z and a friction coefficient μ . This relationship is expressed in the following equation (10). The notation "*" is used to denote pure horizontal forces, such as longitudinal forces when there are no lateral forces, and vice versa. The interplay between these forces will be discussed later in the Forces Correlation section (Section 2.7.2).

$$F_l^* = \mu_l(s, F_z) \cdot F_z, \quad F_{lat}^* = \mu_{lat}(\alpha, F_z) \cdot F_z. \quad (10)$$

2.7.1 Magic Formula

Identifying the friction coefficient is a challenging task as it depends on various factors such as road texture, humidity, tire temperature, and more. In this study, the research is based on typical tires labeled as "r25b" on dry asphalt. Due to the limitations of testing tire friction at EPFL, the study relies on data provided by AMZ [4].

The available data provides the longitudinal friction coefficient as a function of the slip ratio (Section 2.6), and the lateral friction coefficient as a function of the slip angle (Section 2.6), considering a specific normal force F_z . The obtained results are presented below.

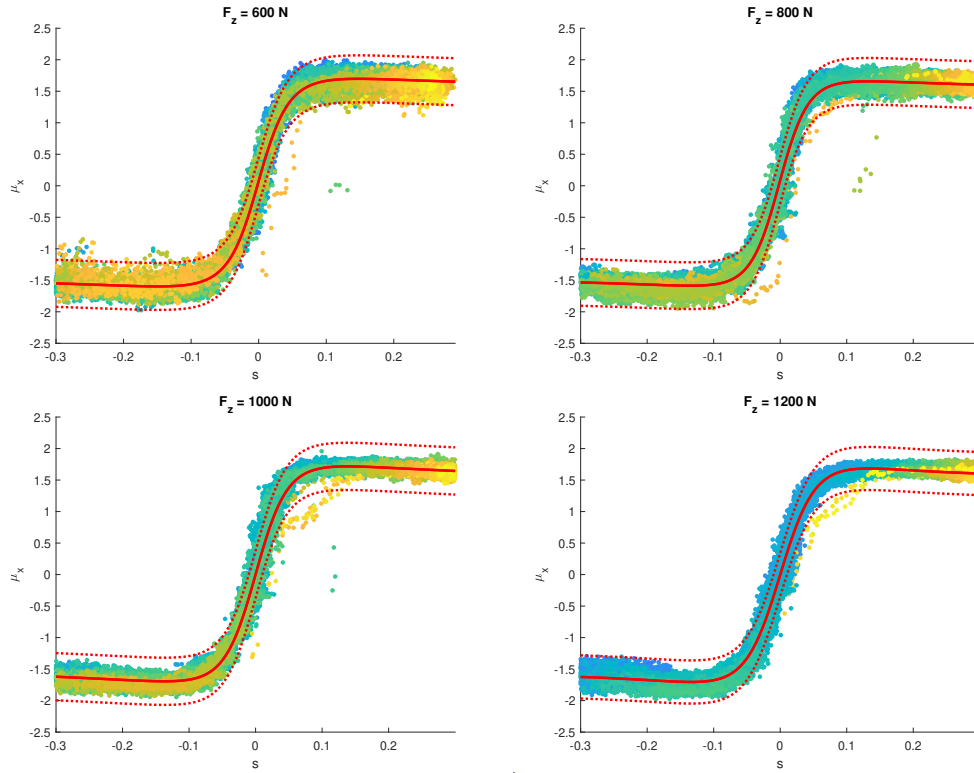


Figure 7: longitudinal friction coefficient in terms of slip ratio

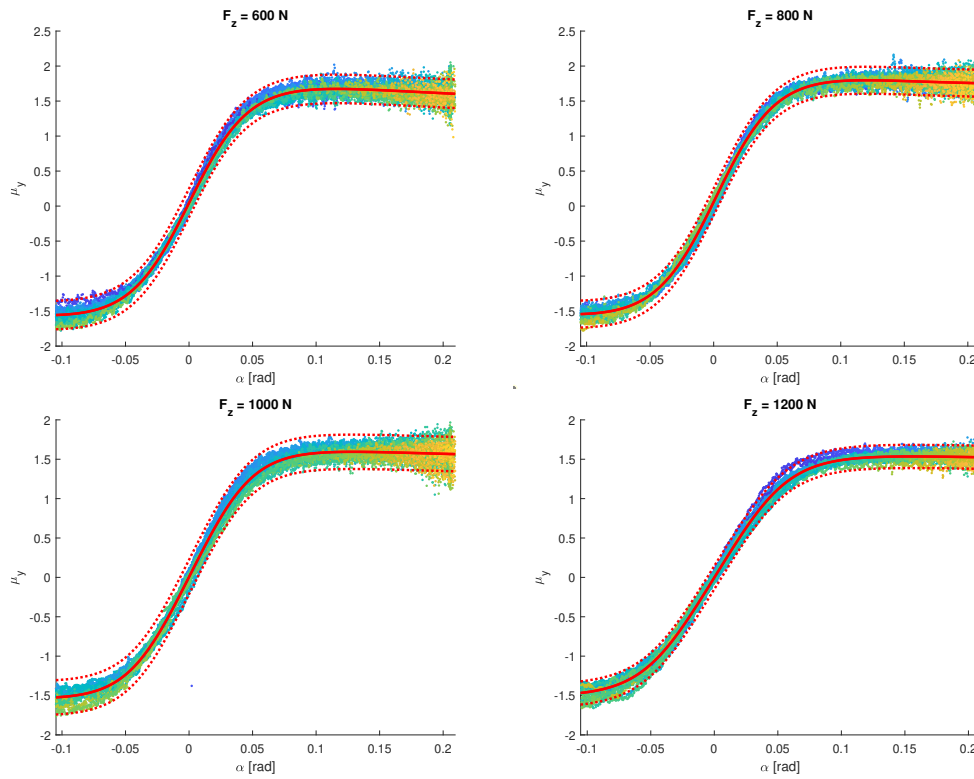


Figure 8: lateral friction coefficient in terms of slip angle

The red curves represented on the data are generated by fitting a parametrization using the widely used magic formula [5], with a 95% confidence interval. This formula, referred to as the magic formula, is the prevailing approach for tire modeling. Despite lacking a direct physical interpretation, this formula (11) provides the best fit to the tire data. The coefficients within the formula are known as the Pacejka coefficients, named after the inventor of the formula.

$$\begin{cases} \mu_l = D_s \sin(C_s \arctan(B_s s - E_s(B_s s - \arctan(B_s s)))) \\ \mu_{lat} = D_\alpha \sin(C_\alpha \arctan(B_\alpha \alpha - E_\alpha(B_\alpha \alpha - \arctan(B_\alpha \alpha)))) \end{cases} \quad (11)$$

To generalize the coefficient identification for any normal force F_z considered in the data, a possible approach is to parameterize each coefficient in terms of F_z . This methodology was discussed in a 2015 research paper titled "Essentials of Vehicle Dynamics" [6]. By eliminating horizontal and vertical shifts from the data, the parameters can be computed according to the equation (12). This enables the estimation of the coefficients for various normal forces, allowing for a more comprehensive tire model.

$$\begin{cases} BCD_s = (b_1 F_z^2 + b_2 F_z) e^{-b_3 F_z} \\ C_s = c_1 \\ D_s = d_1 F_z + d_2 \\ E_s = e_1 F_z^2 + e_2 F_z + e_3 \end{cases} \quad \& \quad \begin{cases} BCD_\alpha = b_1 \sin(2 \arctan(F_z/b_2)) \\ C_\alpha = c_1 \\ D_\alpha = d_1 F_z + d_2 \\ E_\alpha = e_1 F_z^2 + e_2 F_z + e_3 \end{cases} \quad (12)$$

The obtained values for the different coefficients are presented in Table 2.

μ_{long}		μ_{lat}	
b_1	$-6.75 \cdot 10^{-6}$	b_1	$3.79 \cdot 10^1$
b_2	$1.35 \cdot 10^{-1}$	b_2	$5.28 \cdot 10^2$
b_3	$1.20 \cdot 10^{-3}$		
c_1	1.86	c_1	1.57
d_1	$1.12 \cdot 10^{-4}$	d_1	$-2.03 \cdot 10^{-4}$
d_2	1.57	d_2	1.77
e_1	$-5.38 \cdot 10^{-6}$	e_1	$-2.24 \cdot 10^{-3}$
e_2	$1.11 \cdot 10^{-2}$	e_2	1.81
e_3	-4.62		

Table 2: Identified coefficients for magic formula fit

Finally, a comprehensive overview of the pure friction coefficient can be calculated based on the slips and the normal force. Utilizing the aforementioned coefficients, the resulting 3D curves are depicted in Figure 9. These curves provide a visualization of the relationship between the friction coefficient, slip, and normal force.

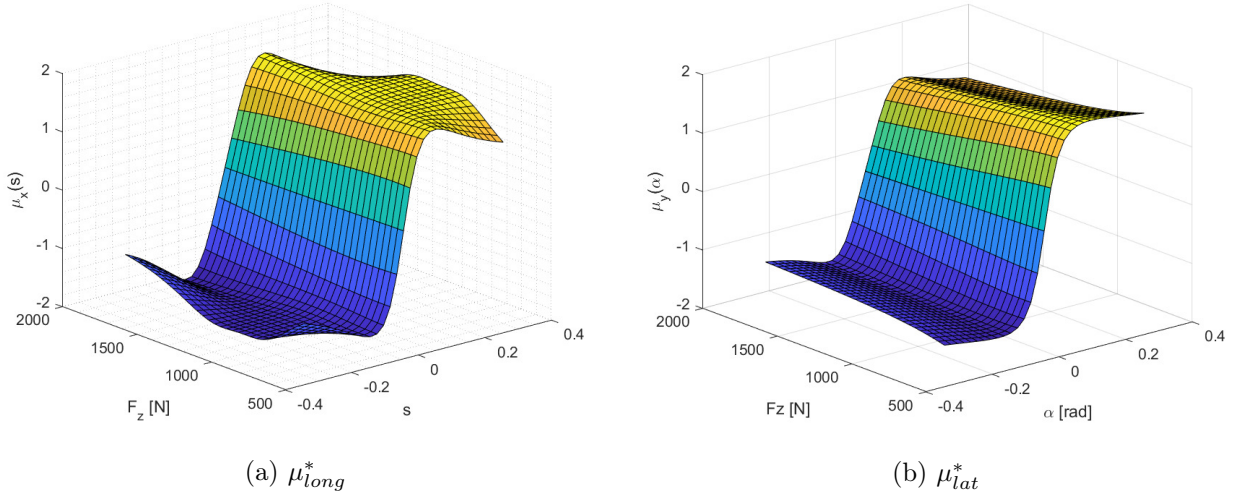


Figure 9: identified friction coefficient in terms of slip and normal force

2.7.2 Forces correlation

Up until this paragraph, only pure horizontal forces have been calculated. There are various methods to compute the correlation between longitudinal and horizontal forces, and two approaches are studied in this document: the traction circle model and the modified Nicolas-Comstock model [7]. These models aim to capture the relationship between longitudinal and horizontal forces in a tire.

Traction circle model : The simplest model is the traction circle, which assumes a small slip angle α that is equivalent to the lateral slip. Therefore, a normalized representation of the horizontal forces can be expressed as shown in equation (13). This model simplifies the correlation between longitudinal and lateral forces by considering a circular relationship.

$$F_{long} = \frac{|s|}{\sqrt{s^2 + \tan^2(\alpha)}} \cdot F_{long}^*, \quad F_{lat} = \frac{|\tan(\alpha)|}{\sqrt{s^2 + \tan^2(\alpha)}} \cdot F_{lat}^*. \quad (13)$$

Modified Nicolas-Comstock model : A more accurate approach for modeling the correlation between horizontal forces is the modified Nicolas-Comstock model [7]. This model takes into account higher slip angles α and provides a better representation of the relationship between longitudinal and lateral forces. The equations describing this correlation are presented below as (14).

$$F_{long} = \frac{F_{long}^* F_{lat}^* s}{\sqrt{s^2 F_{lat}^{*2} + F_{long}^{*2} \tan^2(\alpha)}} \cdot \frac{\sqrt{s^2 K_\alpha^2 + (1 - |s|)^2 \cos^2(\alpha) F_{long}^{*2}}}{s K_\alpha}, \quad (14)$$

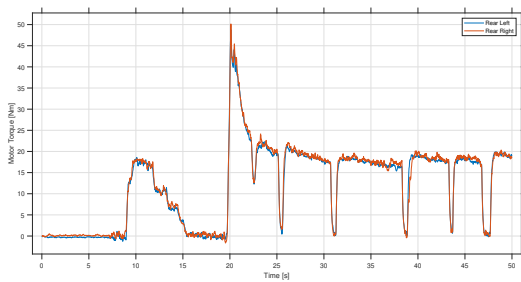
$$F_{lat} = \frac{\text{sign}(s) F_{long}^* F_{lat}^* \tan(\alpha)}{\sqrt{s^2 F_{lat}^{*2} + F_{long}^{*2} \tan^2(\alpha)}} \cdot \frac{\sqrt{\sin^2(\alpha) K_s^2 + (1 - |s|)^2 \cos^2(\alpha) F_{lat}^{*2}}}{\sin(\alpha) K_s}.$$

In the equations, K_s and K_α represent the longitudinal and lateral stiffness, respectively. These stiffness values can be computed as $K_s = BCD_s \cdot F_z$ and $K_\alpha = BCD_y \cdot F_z$. These stiffness coefficients determine the relationship between the applied forces and the resulting tire deflection in the longitudinal and lateral directions.

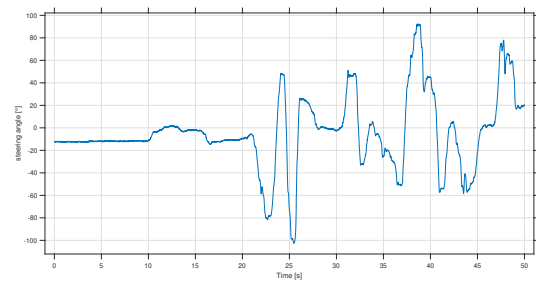
2.8 Simulink validation

In order to assess the validity of the highly nonlinear model described earlier in this section, a Simulink model is implemented, incorporating all the equations discussed. The details of this model can be found in the appendix B. To evaluate the model's performance, a comparison is made between the simulation outputs and the filtered measurements. The available measurements include longitudinal and lateral accelerations, yaw rate, and rear wheel encoders (since the tests were conducted on the LRT3 car, which is a 2WD vehicle).

The inputs for the simulation, as shown in Figure 10, consist of motor torques and the steering angle of the drive wheel. For these tests, real data collected during an endurance race in FSG (Formula Student Germany) during the summer of 2022 are utilized and presented below. This real-world data serves as a basis for evaluating the model's performance and its ability to capture the dynamics of the vehicle.



(a) applied motor torques



(b) applied steering angle

Figure 10: Inputs of the simulation based on real filtered measurements

Two sets of results are examined in this study. The first set pertains to the simulation conducted prior to incorporating the internal motor frictions and brake forces into the model. The results obtained from this initial simulation are presented in Figure 11. These results provide insights into the performance of the model without considering the internal motor frictions and brake forces, allowing for a comparison with the measured data.

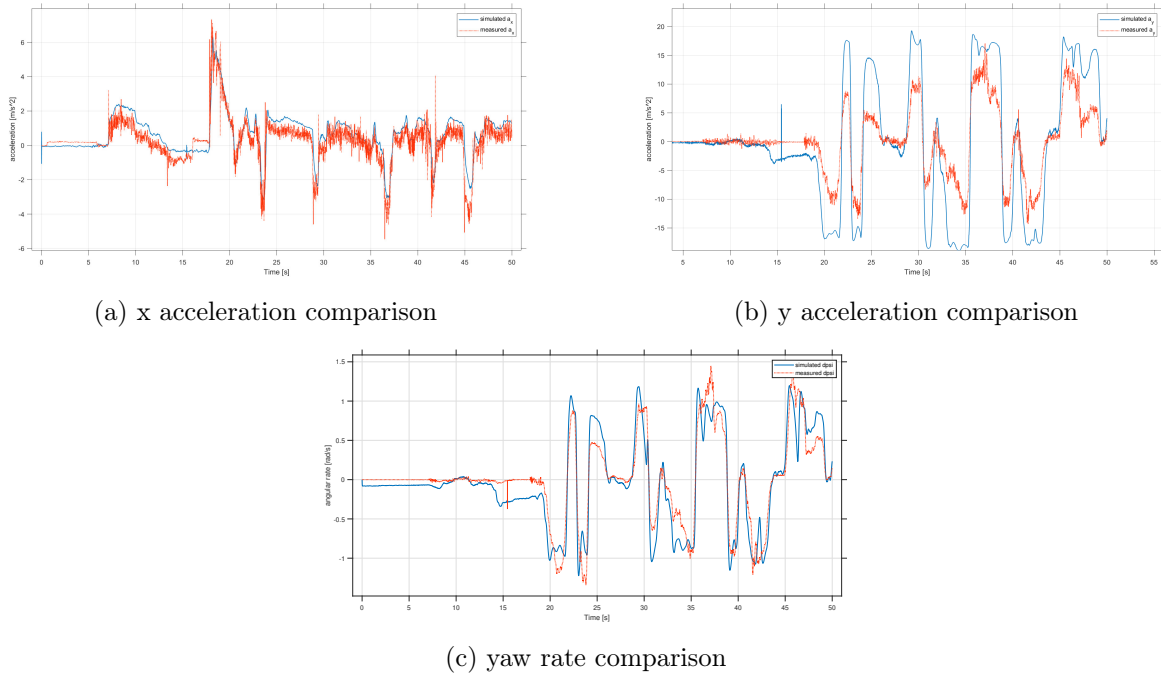


Figure 11: INS validations without internal frictions model

Based on the obtained results, the longitudinal dynamics of the moving car in the simulation appear to be correct. However, a noticeable bias is observed in the x acceleration, which could potentially be attributed to sensor bias. On the other hand, when the car is stationary (around 18 seconds), it becomes evident that the proposed model lacks the incorporation of nonlinear dynamics.

In terms of lateral dynamics, the simulation overestimates the acceleration. The exact source of this discrepancy is challenging to determine solely based on this test. Fortunately, motor speed measurements are also available, allowing for a comparison between the simulation and the measured data. These comparisons are presented in the subsequent plots (Figure 12a), which provide further insights into the agreement between the simulation and the actual measurements.

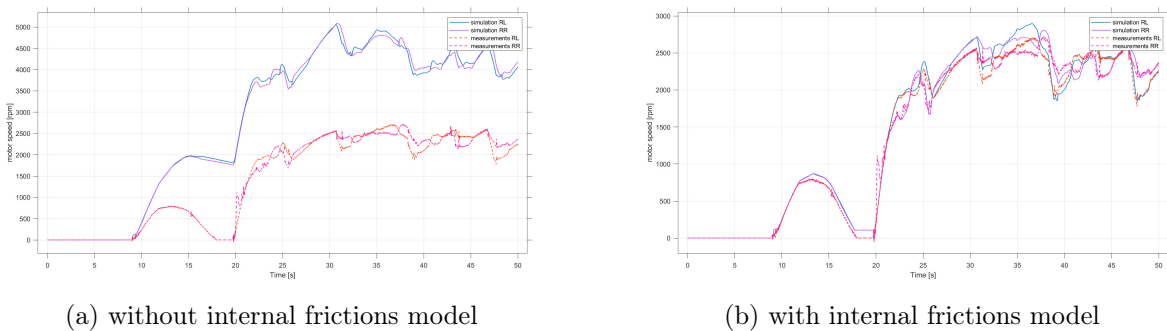
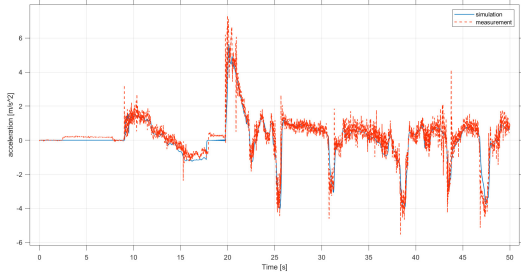


Figure 12: motors speed validations

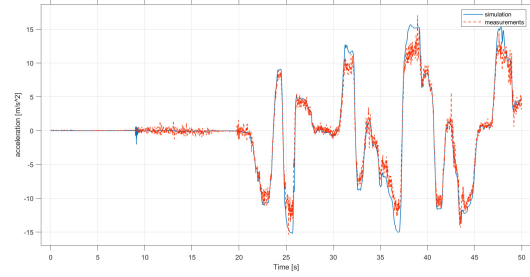
Following the identification of the missing internal motor frictions in the model, further tests were conducted with these additions incorporated (Figure 12b and 13). The results obtained from these tests show significant improvements compared to the previous simulation. The lateral dynamics of the

vehicle are now almost validated, and even the longitudinal dynamics have been enhanced.

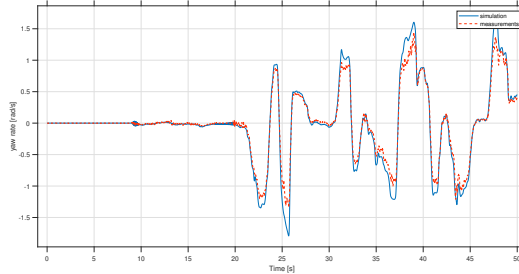
Based on these improved results, the parameters used in this simulation can be deemed suitable for the remainder of the project. They provide a reliable basis for further analysis and development.



(a) x acceleration comparison



(b) y acceleration comparison



(c) yaw rate comparison

Figure 13: INS validations with internal frictions model

2.9 Further improvements

Indeed, the validations conducted so far need to be extended to the new 4-wheel drive car, LRT4. However, it is evident that the current parametrization of the model is not perfect. In order to address this issue and achieve a better parametrization, one possible approach is to formulate an optimization problem using Monte Carlo simulations.

By employing Monte Carlo simulations within an optimization framework, it becomes possible to iteratively refine the model's parameters and improve its accuracy. This approach involves generating random samples of parameter values, running simulations using these values, and comparing the simulation results with experimental data or validated models. Through an iterative process of adjusting parameter values based on the simulation outcomes, the optimization algorithm can converge towards a set of optimized parameters that yield the best match between the model and real-world observations.

By utilizing Monte Carlo simulations and optimization techniques, it becomes feasible to enhance the parametrization of the model and achieve a more accurate representation of the vehicle's dynamics. This iterative process allows for continuous refinement, taking into account the variability and uncertainties inherent in real-world systems.

3 State estimation

In this section, the primary objective is to obtain a highly accurate estimation of the vehicle's state in order to facilitate optimal control. The key quantities to be estimated include the normal forces and slip ratios for traction control purposes. Additionally, the vehicle's velocity is also required for tasks such as torque vectoring and driverless division. While no single sensor can provide velocity measurements with a refresh rate of 100Hz, there are several other sensors available on the vehicle that can be utilized.

These sensors include an Inertial Navigation System (INS), motor current sensors, motor speed encoders, and brake pressure sensors. Each of these sensors has its own measurement accuracy and update rate ($T_s = 10$ ms). To achieve the best possible estimation, various estimation techniques, including the Mixed Kalman Filter (MKF), are employed.

3.1 Sensors

The sensors available on the vehicle are describe here with there models. There is an INS and a motor speed encoder for each motor. Additionally, the motor torques are available from currents and the brake torque is calculated from the brake pressures. The refresh rate is $T_s = 0.01s$.

3.1.1 steering encoder

The wheel steer angle is measured using an absolute encoder, which provides a highly accurate measurement. It is assumed that the measurements from the encoder are free from any noise or disturbances.

$$\tilde{\gamma} = \gamma. \quad (15)$$

3.1.2 INS

The Inertial Navigation System (INS) utilizes selected sensors such as the horizontal accelerometers and the gyroscope along the z-axis. It is known that the accelerometer measurements are affected by bias, while the gyroscope measurements are assumed to be unbiased. The sensor models for the INS can be expressed as shown in equation (16).

$$\begin{cases} \tilde{a}_x = a_x + b_{a_x} + \eta_{a_x} \\ \tilde{a}_y = a_y + b_{a_y} + \eta_{a_y} \\ \tilde{\dot{\psi}} = \dot{\psi} + \eta_{\dot{\psi}} \end{cases} \quad (16)$$

3.1.3 Motors speed encoders

The motor speeds are accurately obtained through the use of encoders, ensuring that there is no bias in this particular model. The encoder model can be represented as shown in equation (17).

$$\tilde{\omega}_{m,i} = \omega_{m,i} + \eta_{\omega_m}. \quad (17)$$

3.1.4 motors currents

The motor current can be easily measured, and the current-to-torque relationship is typically provided by the manufacturer. Since this measurement is assumed to be unbiased, the model of the current sensors can be represented as shown in equation (18).

$$\tilde{i}_{m,i} = i_{m,i} + \eta_{i_m}. \quad (18)$$

3.1.5 Brake pressures

The brake pressures are measured using sensors, with separate sensors for the front and rear pressures. These measurements can be represented by the model given in equation (19).

$$\tilde{P}_{b,j} = P_{b,j} + \eta_{P_b}. \quad (19)$$

3.2 Preliminary transformations

In this section, various transformations are performed on the sensor values to obtain usable measurements. These transformations include unit conversions and simple calculations. Examples of such transformations are the conversion from motor current to wheel torque and the conversion from motor speed to wheel speed. These transformations are described in detail in this section.

3.2.1 Wheels steers

To determine the steer angle of each wheel, δ_L and δ_R , the measurements from the steering wheel encoder are utilized. The equation (4) described in the Vehicle dynamics section (Section 2) is employed for this purpose, and it is restated below as (20). This equation allows the computation of the wheel steer angles based on the measurements obtained from the steering wheel encoder.

$$\tilde{\delta}_{FL}(k) = \begin{cases} k_{in}\tilde{\gamma}(k) & , \text{if } \tilde{\gamma}(k) < 0 \\ k_{out}\tilde{\gamma}(k) & , \text{otherwise} \end{cases}, \quad \tilde{\delta}_{FR}(k) = \begin{cases} k_{out}\tilde{\gamma}(k) & , \text{if } \tilde{\gamma}(k) < 0 \\ k_{in}\tilde{\gamma}(k) & , \text{otherwise} \end{cases}. \quad (20)$$

3.2.2 Wheels torques

The motor torques are calculated directly from the motor currents using the specifications provided by the manufacturer, and their accuracy is verified by the team members. A lookup table (Table 3) is utilized to establish the relationship between the motor currents and the corresponding motor torques. This lookup table provides the necessary data to convert the measured motor currents into corresponding torque values.

i_i [A]	0	25	50	75	100	125	150	160	170	180	190	200
$\tau_{m,i}$ [Nm]	0	17.5	30	42.5	53	65	76	79	82	86	88	90

Table 3: Lookup table from motor current i_i to motor torque $\tau_{m,i}$

The gear ratio η is taken into account to determine the effective motor torque transmitted to the wheels. By multiplying the motor torque by the gear ratio, the torque value is adjusted to reflect the actual torque applied at the wheels.

$$\tilde{\tau}_{w,i}(k) = \tilde{\tau}_{m,i}(k) \cdot \eta. \quad (21)$$

3.2.3 Wheels speeds

To obtain the wheel speed, the value obtained from the motor speed encoder is divided by the gear ratio η . This division accounts for the gear reduction or amplification provided by the gear system, resulting in the corresponding wheel speed.

$$\tilde{\omega}_{w,i}(k) = \frac{\tilde{\omega}_{m,i}(k)}{\eta}. \quad (22)$$

3.2.4 Wheels accelerations

To compute the wheel accelerations while mitigating the effects of high-frequency noise, an average derivative is calculated over a certain number of sampling periods. This approach helps to smooth out the acceleration measurements and provides a more reliable estimate. The average derivative is computed using the following equation:

$$\tilde{\dot{\omega}}_i(k) = \frac{\tilde{\omega}_i(k) - \tilde{\omega}_i(k - 20)}{20 \cdot T_s}. \quad (23)$$

This approach helps to filter out high-frequency noise and fluctuations in the wheel speed measurements, allowing for a more accurate estimation of the wheel accelerations, which are important for analyzing the dynamics of the vehicle.

3.3 General state estimation

After the preliminary transformations, several measurements are computed based on the transformed sensor values and the previous state estimate. These measurements include the calculation of normal forces, longitudinal forces and slip ratios.

3.3.1 Normal forces

The normal forces are determined based on the latest state estimation. By applying the equation (5) provided in the Vehicle dynamics section (Section 2), the normal forces can be computed using the following expression (24):

$$\begin{cases} \hat{F}_{z,FL}(k) = \frac{mgl_r}{2l} + \frac{mz_{CG}}{2a} \hat{a}_y(k-1) - \frac{mz_{CG}}{2l} \hat{a}_x(k-1) + F_{lift}(\hat{v}_x(k-1)) \\ \hat{F}_{z,FR}(k) = \frac{mgl_r}{2l} - \frac{mz_{CG}}{2a} \hat{a}_y(k-1) - \frac{mz_{CG}}{2l} \hat{a}_x(k-1) + F_{lift}(\hat{v}_x(k-1)) \\ \hat{F}_{z,RL}(k) = \frac{mgl_f}{2l} + \frac{mz_{CG}}{2b} \hat{a}_y(k-1) + \frac{mz_{CG}}{2l} \hat{a}_x(k-1) + F_{lift}(\hat{v}_x(k-1)) \\ \hat{F}_{z,RR}(k) = \frac{mgl_f}{2l} - \frac{mz_{CG}}{2b} \hat{a}_y(k-1) + \frac{mz_{CG}}{2l} \hat{a}_x(k-1) + F_{lift}(\hat{v}_x(k-1)) \end{cases}. \quad (24)$$

3.3.2 Longitudinal forces

The calculation of longitudinal forces incorporates a non-linear relationship to account for the stop case. This case is identified when both the wheel speed ω and its derivative $\dot{\omega}$ are zero. To accommodate measurement noise, a small threshold ϵ is chosen. In this scenario, the longitudinal force is expected to be zero. For all other cases, the difference between the applied torque, friction torque, and wheel inertia is calculated, as expressed in equation (25).

$$\tilde{F}_{l,i}(k) = \begin{cases} 0 & \text{if } |\tilde{\omega}_i| < \epsilon \text{ and } |\tilde{\dot{\omega}}_i| < \epsilon \\ \frac{1}{R_w}(\tilde{\tau}_i(k) - \tilde{\tau}_{f,i}(k) - I_w \tilde{\dot{\omega}}_i(k)) & \text{otherwise} \end{cases}. \quad (25)$$

This approach ensures that the longitudinal force is accurately determined, considering different operating conditions such as wheel speed and the effects of torque and friction. By considering the stop case and accounting for measurement noise, a more comprehensive and reliable estimation of longitudinal forces can be obtained.

3.3.3 slips ratios

To compute the slip ratios, several approaches have been studied, and all of them are presented here. However, the final choice is to estimate them using the Kalman filter, as explained in the next section (Section 3.4). In this paragraph, the initial ideas for slip ratio estimation are discussed.

The first proposition involves a linear approximation. With a known longitudinal force and normal force, it is possible to invert the magic formula in its linear domain ($s < 0.05$). The equation representing this relationship is presented below as (26).

$$\hat{s}_i(k) = \frac{\tilde{F}_{l,i}(k)}{\hat{F}_{z,i}(k)BCD_s(\hat{F}_{z,i}(k))}. \quad (26)$$

This linear approximation provides a straightforward method for estimating slip ratios based on known longitudinal and normal forces. By inverting the magic formula in the linear domain, a reasonably accurate estimation can be obtained within a specific range of slip ratios. However, it should be noted that this approach is limited to the linear range and may not capture the full complexity of the tire behavior at higher slip ratios.

Another proposition is to invert the magic formula in a bijective domain ($s < 0.14$) as shown in Figure 15. By observing that the longitudinal friction coefficient μ_l has a low dependency on the normal force F_z , it is possible to create a lookup table that facilitates the inversion of the magic formula. This approach is summarized in equation (27), and the inversion graph is presented thereafter (Figure 14).

$$\hat{s}_i(k) = \mu_l^{-1} \left(\frac{\tilde{F}_{l,i}(k)}{\hat{F}_{z,i}(k)} \right). \quad (27)$$

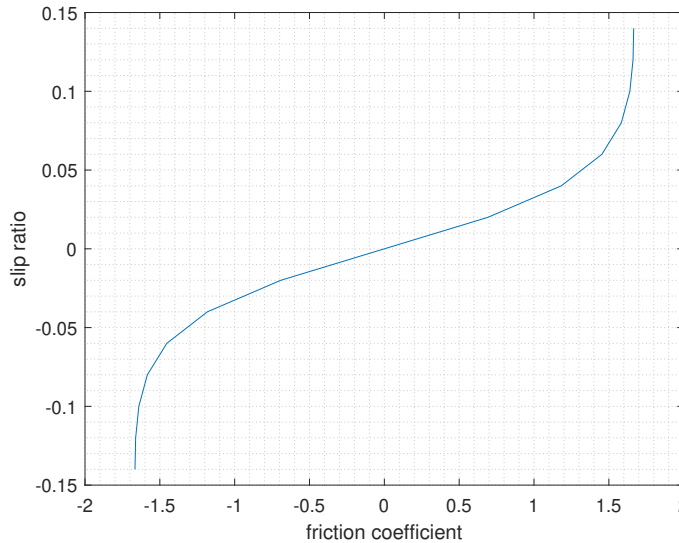


Figure 14: Lookup table used in the bijective domain

By creating a lookup table based on this observation, it becomes easier to estimate the slip ratios by inverting the magic formula within the specified range. However, it is important to note that this

method is still limited to a certain range of slip ratios (Figure 15) and may not accurately capture the tire behavior outside of this range.

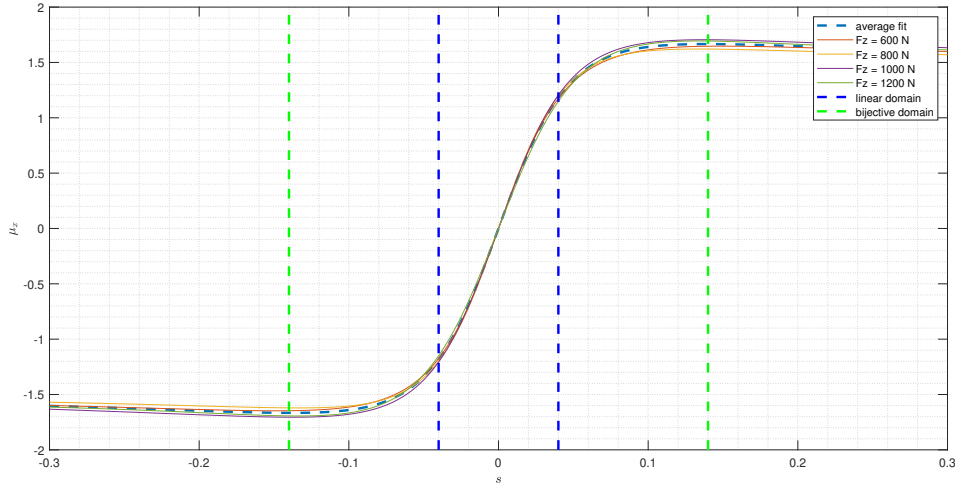


Figure 15: Graph presented the different slip estimation method domains

In order to estimate the slip ratios in the entire domain, it is necessary to utilize the definition explained in the Vehicle dynamics section (Section 2). However, this definition is not applicable when the velocity is null. To address this limitation and enable slip estimation across all operating conditions, the decision has been made to include the slip ratios in the Kalman filter, which will be discussed in the next section (Section 3.4). By incorporating the slip ratios into the estimation process, it becomes possible to overcome the challenges associated with null velocity and obtain more accurate slip estimations.

3.4 Kalman filter

In this section, a Mixed Kalman filter is introduced to prioritize the estimation of vehicle velocity and slip ratios. Due to the availability of multiple sensors with known accuracy and the assumption of white Gaussian noise, a Kalman filter can be employed for state estimation. However, to achieve the best possible estimation, a combination of different Kalman filter variants is utilized: the Extended Kalman Filter (EKF), Linear Kalman Filter (LKF), and Unscented Kalman Filter (UKF). This estimator development is based on a research of the autonomous Systems Lab of ETH Zürich [8]. A Schema presented the different parts of this MKF is presented in Figure 16.

A previous version of the Mixed Kalman Filter (MKF), which did not include slip ratios in the state estimation, was tested earlier and its details are presented in Appendix D. This earlier version of the MKF focused on estimating other state variables without explicitly incorporating slip ratios into the estimation process. The purpose of testing this previous version was to assess its performance and evaluate its effectiveness in estimating the desired variables. The results and findings from this testing phase are documented in the respective appendix.

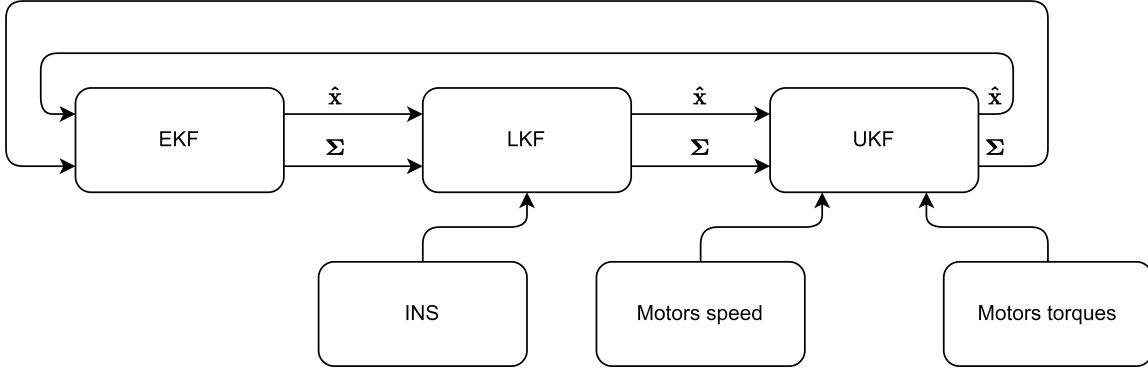


Figure 16: Schema of the MKF design

The EKF is suitable for propagating the nonlinear dynamics of the system, while employing a linear approximation for covariance propagation. This allows for efficient estimation of the system's nonlinear behavior. The LKF, on the other hand, facilitates easy comparison between INS measurements and the actual estimate using a linear relationship. This linear nature simplifies the analysis of the sensor measurements and their integration into the estimation process.

In cases where a highly nonlinear relationship exists between the state and measurements, the UKF is implemented. The UKF is particularly useful when dealing with complex nonlinearities, such as those encountered when estimating wheel torques and speeds. By employing the UKF, more accurate and reliable estimations can be obtained in highly nonlinear scenarios.

By combining the strengths of these different Kalman filter variants, the Mixed Kalman filter provides a comprehensive and robust approach for estimating the vehicle velocity and slip ratios, leveraging the capabilities of each filter variant based on the specific characteristics of the system dynamics and sensor measurements.

The state of this MKF is the following.

$$\mathbf{x} = [v_x \ v_y \ a_x \ a_y \ \dot{\psi} \ s_{FL} \ s_{FR} \ s_{RL} \ s_{RR}]^T. \quad (28)$$

3.4.1 Extended Kalman filter

The Extended Kalman Filter (EKF) is employed to propagate the dynamics of the system using its nonlinear equations. Although the dynamics are inherently nonlinear, a linearization step is performed to propagate the covariance of the estimation. The dynamics, along with their associated uncertainties, are expressed in the equation (29). The navigation equations used to propagate the estimated state are provided on the right-hand side of the equation. These equations take into account the system dynamics and incorporate the uncertainties to update the state estimate as the system evolves over time.

$$\dot{\mathbf{x}} = \begin{cases} \dot{v}_x = a_x + v_y \dot{\psi} + \xi \dot{v}_x \\ \dot{v}_y = a_y - v_x \dot{\psi} + \xi \dot{v}_y \\ \dot{a}_x = \xi \dot{a}_x \\ \dot{a}_y = \xi \dot{a}_y \\ \ddot{\psi} = \xi \ddot{\psi} \\ \dot{s}_i = \xi \dot{s}_i \end{cases} \quad ; i = 1, \dots, 4 \quad \Rightarrow \quad \hat{\dot{\mathbf{x}}} = \begin{cases} \hat{\dot{v}}_x = \hat{a}_x + \hat{v}_y \hat{\dot{\psi}} \\ \hat{\dot{v}}_y = \hat{a}_y - \hat{v}_x \hat{\dot{\psi}} \\ \hat{\dot{a}}_x = 0 \\ \hat{\dot{a}}_y = 0 \\ \hat{\ddot{\psi}} = 0 \\ \hat{\dot{s}}_i = 0 \end{cases} \quad ; i = 1, \dots, 4 \quad (29)$$

The linearized Kalman matrices are computed as shown in equation (30). The matrices capture the linearization of the system dynamics and measurements around the current state estimate, allowing for efficient estimation updates and covariance propagation in the EKF algorithm.

$$\mathbf{A}_{\text{ekf}} = \begin{bmatrix} 0 & \hat{\psi} & 1 & 0 & \hat{v}_y & \mathbf{0}_{2 \times 4} \\ -\hat{\psi} & 0 & 0 & 1 & -\hat{v}_x & \mathbf{0}_{7 \times 4} \\ & & \mathbf{0}_{7 \times 5} & & & \mathbf{0}_{7 \times 4} \end{bmatrix}, \quad \mathbf{B}_{\text{ekf}} = \mathbf{I}_9, \quad \mathbf{Q}_{\text{ekf}} = \text{diag}([\sigma_{\xi \dot{v}_x}^2, \sigma_{\xi \dot{v}_y}^2, \sigma_{\xi \dot{a}_x}^2, \sigma_{\xi \dot{a}_y}^2, \sigma_{\xi \ddot{\psi}}^2, \sigma_{\xi \dot{s}_i}^2 (x4)]). \quad (30)$$

The propagation step in the Extended Kalman Filter follows the principles of the EKF theory. The equations used for propagation (31) are derived from the navigation systems course notes by J. Le Ny in Polytechnique Montréal [9], [10].

$$\begin{cases} \hat{\mathbf{x}}_{\mathbf{k}+1} = \hat{\mathbf{x}}_{\mathbf{k}} + \hat{\dot{\mathbf{x}}}_{\mathbf{k}} \cdot T_s \\ \Sigma_{\mathbf{k}+1} = \Phi_{\mathbf{k}} \Sigma_{\mathbf{k}} \Phi_{\mathbf{k}}^T + \mathbf{N} \end{cases} \quad \text{with} \quad \Phi_{\mathbf{k}} = e^{\mathbf{A}_{\mathbf{k}} T_s} \quad \text{and} \quad \mathbf{N} \simeq \mathbf{B} \mathbf{Q} \mathbf{B}^T T_s. \quad (31)$$

These equations are used to update the state estimate and propagate the covariance matrix. They take into account the system dynamics and the uncertainty in the state estimate. By integrating these equations over time, the EKF is able to estimate the state of the system with improved accuracy.

It's important to note that these equations are specific to the EKF algorithm and are designed to handle non-linear dynamics by linearizing them around the current state estimate.

3.4.2 Linear Kalman filter

In the implementation of the Linear Kalman Filter (LKF), the goal is to compare the Inertial Navigation System (INS) measurements (Section 3.1.2) with the actual state estimation. In this case, three measurements are performed, and they have a linear relation with the state. Therefore, the observation matrices used in the LKF are as follows:

$$\mathbf{C}_{\text{lkf}} = [\mathbf{0}_{3 \times 2} \quad \mathbf{I}_3 \quad \mathbf{0}_{3 \times 4}], \quad \mathbf{D}_{\text{lkf}} = \mathbf{I}_3, \quad \mathbf{N}_{\text{lkf}} = \text{diag}([\sigma_{\eta_{a_x}}^2, \sigma_{\eta_{a_y}}^2, \sigma_{\eta_{\dot{\psi}}}^2]). \quad (32)$$

Based on the Kalman filter theory, the Kalman gain is computed, and the estimate update is performed in the Linear Kalman Filter (LKF). The equations used for the estimate update are shown in equation (33).

$$\begin{cases} \hat{\mathbf{x}}_{\mathbf{k}}^+ = \hat{\mathbf{x}}_{\mathbf{k}}^- + \mathbf{K}_{\mathbf{k}}(\tilde{\mathbf{y}}_{\mathbf{k}} - \mathbf{C}\hat{\mathbf{x}}_{\mathbf{k}}^-) \\ \Sigma_{\mathbf{k}}^+ = (\mathbf{I}_9 - \mathbf{K}_{\mathbf{k}}\mathbf{C})\Sigma_{\mathbf{k}}^- \end{cases} \quad \text{with} \quad \mathbf{K}_{\mathbf{k}} = \Sigma_{\mathbf{k}}^- \mathbf{C}^T (\mathbf{C}\Sigma_{\mathbf{k}}^- \mathbf{C}^T + \mathbf{D}\mathbf{N}\mathbf{D}^T)^{-1}. \quad (33)$$

3.4.3 Unscented Kalman filter

The Unscented Kalman Filter (UKF) is used in this context because the Extended Kalman Filter (EKF) requires linearization, which may not provide accurate results for highly nonlinear relationships. The UKF offers an alternative approach by creating a set of sigma points that are distributed according to the covariance matrix. These sigma points are then propagated through the nonlinear function, allowing for a more accurate estimation of the covariance.

The basic steps of the UKF are based on a article from E.Wan and M.Rudolph [11] and can be summarized as follows:

1st step (create points) : To create the sigma points in the Unscented Kalman Filter (UKF), the distribution of points is determined based on the current state estimate and its associated covariance matrix. One common method to calculate the sigma points is by using the following equation:

$$\mathcal{X}_j = \begin{cases} \hat{\mathbf{x}}_{\mathbf{k}} & ; j = 0 \\ \hat{\mathbf{x}}_{\mathbf{k}} + (\sqrt{L + \lambda} \cdot \text{chol}(\underline{\Sigma}_{\mathbf{k}}^-)^T)_j & ; j = 1, \dots, L \\ \hat{\mathbf{x}}_{\mathbf{k}} - (\sqrt{L + \lambda} \cdot \text{chol}(\underline{\Sigma}_{\mathbf{k}}^-)^T)_j & ; j = L + 1, \dots, 2L \end{cases} . \quad (34)$$

2nd step (weights) : In the Unscented Kalman Filter (UKF), each sigma point is associated with a weight that determines its contribution to the estimation process. The weights are calculated based on the number of sigma points ($2L+1$) and a scaling factor (λ) that depends on the specific implementation.

The weight associated with the mean estimate is denoted as $W_{m,0}$, and the weights for the remaining sigma points are denoted as $W_{m,j}$ and $W_{c,j}$. These weights are computed as follows:

$$W_{m,j} = \begin{cases} \frac{\lambda}{L+\lambda} & ; j = 0 \\ \frac{1}{2*(L+\lambda)} & ; otherwise \end{cases}, \quad W_{c,j} = \begin{cases} \frac{\lambda}{L+\lambda} + 1 - \alpha^2 + \beta & ; j = 0 \\ \frac{1}{2*(L+\lambda)} & ; otherwise \end{cases} . \quad (35)$$

3rd step (observations) : In the third step of the Unscented Kalman Filter (UKF), the observation function is applied to each of the sigma points created in the previous step. The observation function represents the relationship between the state variables and the measurements obtained from sensors.

$$\mathcal{Y}_j = h(\mathcal{X}_j). \quad (36)$$

4th step (averaging) : In the fourth step of the Unscented Kalman Filter (UKF), a weighted average of the predicted measurements from the previous step is computed. This average, denoted as $\hat{y}$, represents the estimated measurement based on the current state estimate and the observation function.

The weighted average is calculated by taking the sum of the predicted measurements multiplied by their corresponding weights. The weights are assigned based on the sigma point weights computed earlier.

In addition to the weighted average, the weighted covariance and cross-covariance of the observations are also computed. These covariance values provide an estimate of the uncertainty associated with the measurements. A summary of these equations is presented below (37).

$$\begin{cases} \hat{y} = \sum_{j=0}^{2L} W_{m,j} \mathcal{Y}_j \\ \Sigma_{yy} = \sum_{j=0}^{2L} W_{c,j} (\mathcal{Y}_j - \hat{y})^2 \\ \Sigma_{xy} = \sum_{j=0}^{2L} W_{c,j} (\mathcal{X}_j - \hat{x})(\mathcal{Y}_j - \hat{y}) \end{cases} \quad (37)$$

5th step (Kalman update) : In the final step, the Kalman gain is computed based on the weighted covariance and cross-covariance of the predicted state and the observations. The Kalman gain determines the amount of weight given to the predicted state and the observed measurements in updating the state estimate.

$$\begin{cases} \hat{\mathbf{x}}_k^+ = \hat{\mathbf{x}}_k^- + \mathbf{K}_k(\tilde{\mathbf{y}}_k - \hat{\mathbf{y}}_k) \\ \Sigma_k^+ = \Sigma_k^- - \mathbf{K}_k \Sigma_{yy,k} \mathbf{K}_k^T \end{cases} \quad \text{with} \quad \mathbf{K}_k = \Sigma_{xy,k} \Sigma_{yy,k}^{-1} \quad (38)$$

Application on the considered problem

In the considered problem, there are two measurements that have non-linear relations with the estimated state. The first measurement is the wheel speed, which is determined from the estimated slip $\hat{s}$ and the longitudinal velocity of the wheel $\hat{v}_l$. This relation between the wheel speed and slip is defined in the vehicle dynamics. However, this definition may not be accurate in low-speed situations.

To improve the accuracy of the slip estimate, a second measurement is used, which is the longitudinal force of the wheel $\tilde{F}_l$. This measurement provides a more accurate estimation of the slip, particularly in the bijective domain where the relationship between slip and force is well-defined. However, it is important to ensure the reliability of this measurement.

For safety reasons, the second measurement of longitudinal force is only used if the previous slip estimate falls within the bijective domain and is sufficiently far away from the boundaries. This ensures that the measurement is reliable and can be used to update the slip estimate with greater accuracy. The equations behind this explanations are shown below.

$$\begin{cases} \tilde{y}_{1,i} = \tilde{\omega}_{w,i} \\ \hat{y}_{1,i} = \frac{\hat{s}_i + 1}{R_w} \hat{v}_{l,i} \end{cases}, \text{ and } \begin{cases} \tilde{y}_{2,i} = \tilde{F}_{l,i} \\ \hat{y}_{2,i} = \hat{F}_{z,i} \mu_l(\hat{s}_i) \end{cases} \quad \text{if } \hat{s}_i(k-1) < 0.05. \quad (39)$$

With,

$$\hat{v}_{l,i} = (\cos \delta_i \quad \sin \delta_i) \cdot \begin{pmatrix} \hat{v}_x - p_{y,i} \hat{\dot{\psi}} \\ \hat{v}_y + p_{x,i} \hat{\dot{\psi}} \end{pmatrix}. \quad (40)$$

By incorporating these two measurements with their non-linear relations to the estimated state, the UKF is able to refine the slip estimate and provide a more accurate representation of the true slip value, taking into account the uncertainties and limitations of the measurements.

3.5 Biases and lateral velocity reset

The figure (Figure 17) illustrates the problems in the current state estimation approach. It is observed that during the initial period when the car is stopped (from 0 to 8 seconds), the accelerometer measurements should ideally be zero since there is no acceleration. However, the estimated accelerations obtained from the current approach show non-zero values, indicating a bias in the accelerometer

measurements.

Similarly, the lateral velocity estimation is solely based on the INS measurements, which may not accurately capture the true lateral velocity of the vehicle. This can lead to inaccuracies in the estimation of slip ratios and other related quantities.

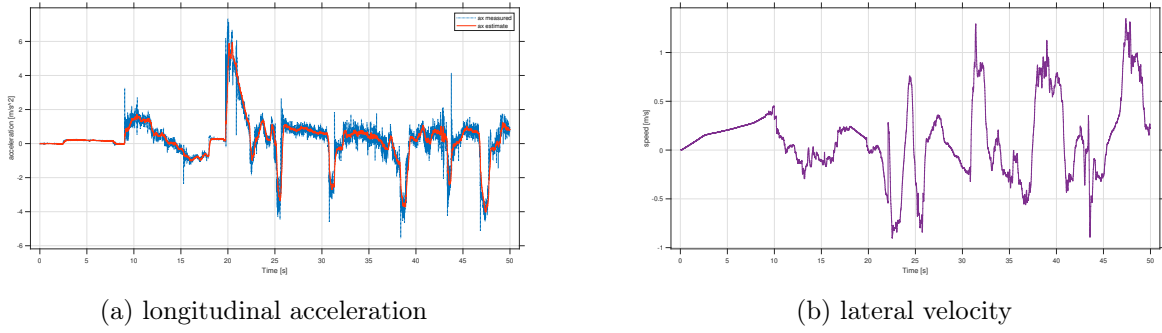


Figure 17: Estimation states which call improvements

The identified issues regarding acceleration biases and the estimation of lateral velocity call for improvements in the state estimation process. One approach to address these challenges is by updating the accelerometer biases when the vehicle is in a stopped condition. This can help mitigate the bias errors and improve the accuracy of acceleration estimation.

Regarding the estimation of lateral velocity, additional conditions can be utilized. In addition to the INS measurements, the lateral velocity estimate can be reset to zero when the vehicle is moving straight for a certain duration, indicated by the condition $|\tilde{\gamma}| < \epsilon$ for a specified number of sampling periods. This resetting mechanism helps maintain accurate estimation of lateral velocity during straight-line driving. To address the update of the covariance during the estimation of lateral velocity, a Linear Kalman Filter (LKF) is implemented.

Based on these considerations, an updated estimation schema, as shown in Figure 18, is implemented. By integrating these improvements into the state estimation process, it is expected that the accuracy and reliability of the estimated state variables, including acceleration and lateral velocity, will be significantly enhanced.

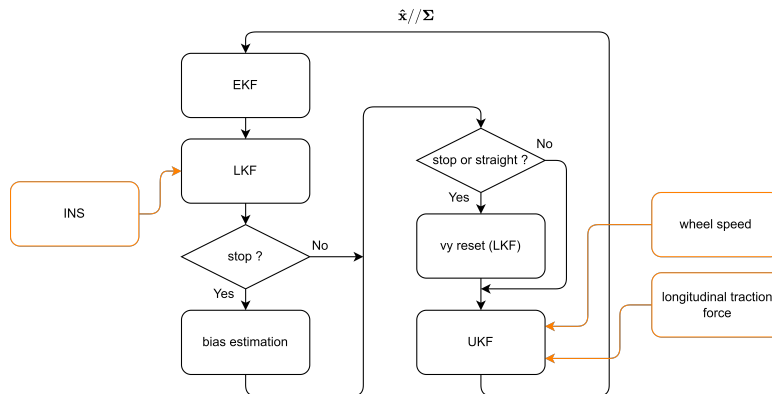


Figure 18: Updated schema of the MKF design

In comparison to the latest estimates presented in Figure 17, the unbiased estimation results are displayed below.

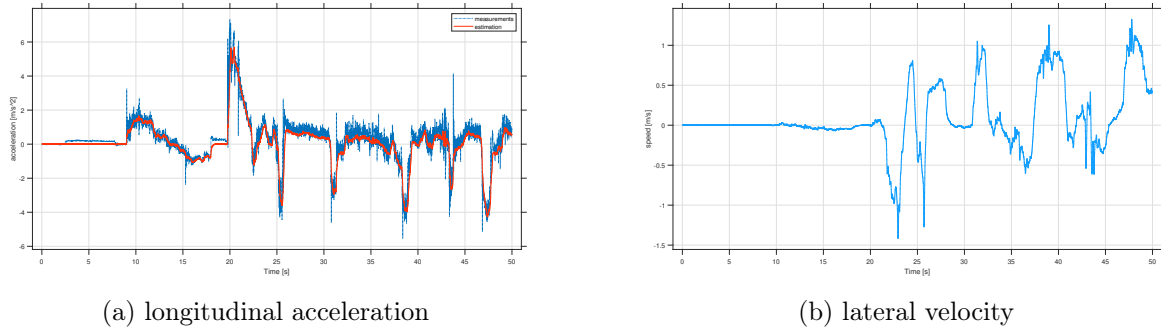


Figure 19: Estimation states with bias estimation

3.6 Testing

In the absence of high-accuracy velocity sensors for direct comparison with the state estimation, the tests conducted using Simulink with existing sensor data serve as a means to evaluate the coherence and filtering performance of the estimation. While it may not be possible to directly validate the state estimation against the ground truth, analyzing the behavior and coherence of the estimated states can provide valuable insights.

Although it may not be possible to directly confirm the accuracy of the estimated states without ground truth data, assessing the coherence of the estimated states can still provide useful information. If the estimated states exhibit smooth and consistent behavior that aligns with the vehicle dynamics and sensor measurements, it indicates that the filtering process is likely functioning properly.

3.6.1 General state estimation

The general transformations described in Section 3.3 are depicted in Figure 20. The figure illustrates the estimation results for normal forces, displayed on the left as shown in Figure 20a. The normal force estimation appears to be highly consistent with the observed car displacements. On the right side of the figure, the differences between the linear and non-linear slip approximations are shown, providing insights into the variations between the two approaches.

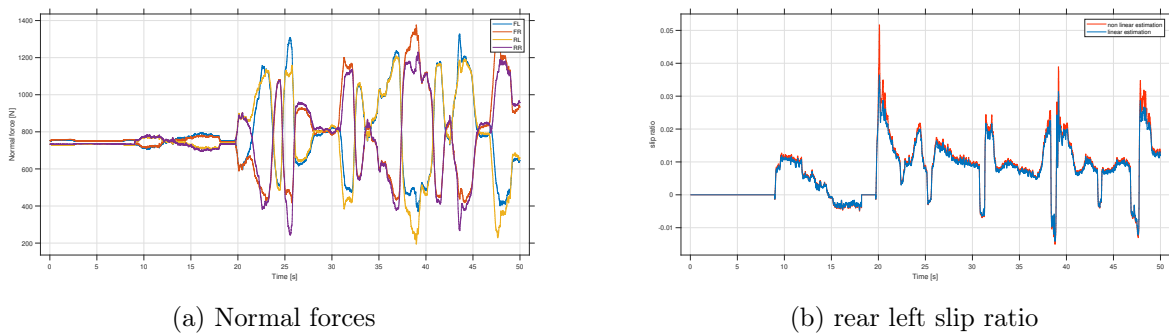


Figure 20: General estimations results

The observations indicate that the linear approximation tends to underestimate the slip ratio compared to the non-linear estimate. This discrepancy has significant implications for traction control systems, which aim to maximize the slip within a defined limit. The accurate estimation of slip is crucial for achieving optimal traction control performance.

3.6.2 Slip estimation

Furthermore, it is evident that even the non-linear approximation does not accurately estimate the slip in the studied example. This is particularly evident when considering the velocity difference between the left and right wheels. In Figure 21a, between 20 and 21 seconds, the left wheel exhibits a significantly higher velocity than the right wheel. This indicates an over-slip situation on the left wheel, which was not detected in the previous slip estimation (Figure 20b).

However, in the new estimation based on the Kalman filter (Figure 21b), this over-slip situation can be effectively detected without sacrificing accuracy within the bijective domain. The Kalman filter-based estimation approach provides improved slip estimation capabilities, enabling the detection of over-slip situations and enhancing the overall accuracy and reliability of the slip estimation process.

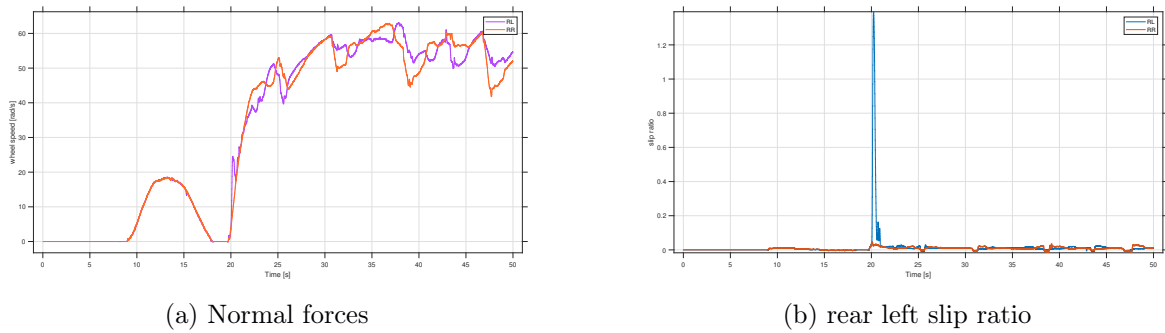


Figure 21: General estimations results

3.6.3 Velocity estimation

Finally, a comparison is made between the previous velocity estimation, which was computed as the mean velocity of the rear wheels, and the new velocity estimation considering the slip ratios. The results are shown in Figure 22.

In the new estimation approach that incorporates slip ratios, it is expected that the velocity estimation will consistently be lower than the wheel speeds, particularly during acceleration. This is because the slip ratios indicate a reduction in effective wheel speed due to slipping. As a result, the velocity estimation accurately reflects the influence of slip on the overall vehicle dynamics.

Comparing the two estimations, it can be observed that the new velocity estimation, provides a more realistic representation of the vehicle's dynamics. This improved estimation is crucial for various control and decision-making algorithms that rely on accurate velocity information.

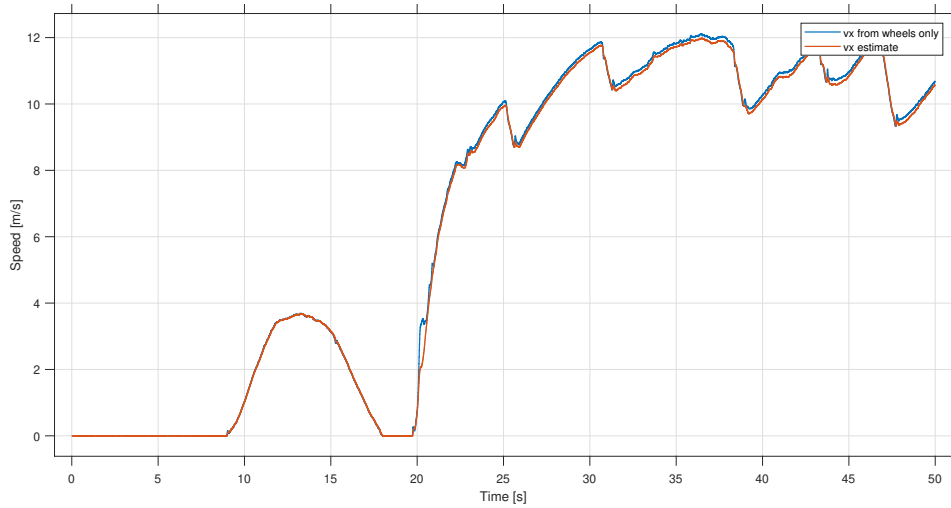


Figure 22: Longitudinal velocity estimation

3.7 Further improvements

The availability of a new high-accuracy velocity sensor is an exciting development for the EPFL racing team. Integrating this sensor into the LRT4 car will provide an opportunity to validate the model-based estimation approach discussed earlier. By comparing the estimations from the sensor with the model-based estimations, the team can assess the accuracy and reliability of the model and potentially refine or improve the estimation algorithms.

In addition to incorporating the new velocity sensor, considering the pitch and roll angles of the car can further enhance the robustness of the state estimation. The pitch angle, which represents the tilt of the car along the lateral axis, and the roll angle, which represents the rotation around the longitudinal axis, can provide valuable information for more accurate acceleration estimation along the desired axis. By taking these angles into account, the estimation algorithms can better compensate for the effects of vehicle orientation on the measured accelerations.

Furthermore, estimating the slip angles can be beneficial for advanced control algorithms. Slip angles, which represent the difference between the actual direction of the wheels and their desired direction, can greatly influence vehicle dynamics, particularly during cornering. By accurately estimating the slip angles, the control algorithms can make more informed decisions to optimize vehicle stability, traction, and overall performance.

In summary, the integration of a high-accuracy velocity sensor, consideration of pitch and yaw angles, and estimation of slip angles can collectively contribute to a more robust and accurate state estimation for further cars. These advancements will not only enhance the performance of control algorithms but also enable more precise and reliable decision-making in various driving conditions.

4 Traction control

Traction control is a crucial aspect of racing car design that focuses on optimizing both safety and performance. It involves the implementation of control algorithms and systems to regulate the torque applied to each wheel of the car, with the goal of preventing excessive wheel slip.

The primary objective of traction control is to maximize traction between the tires and the road surface. When a racing car accelerates or decelerates, there is a limit to how much torque can be applied to the wheels before they start to lose traction and slip. Excessive wheel slip can lead to reduced acceleration, compromised handling, and even loss of control.

4.1 Implementation

The primary objective of a traction control system is to track optimal torques for each wheels based on the driver's commands which cannot be exceeded. To achieve this, the system focuses on controlling the slip ratios of the wheels. By computing an optimal slip ratio based on the current state of the vehicle, the traction controller aims to track this optimal value.

Controlling the slip ratios provides an efficient approach to traction control. The system continuously monitors the state of the car and adjusts the torque distribution to maintain the desired slip ratio. This allows the controller to optimize the traction between the tires and the road surface.

The figure below (Figure 23) provides a visual summary of these explanations, illustrating the concept of controlling slip ratios in the traction control system.

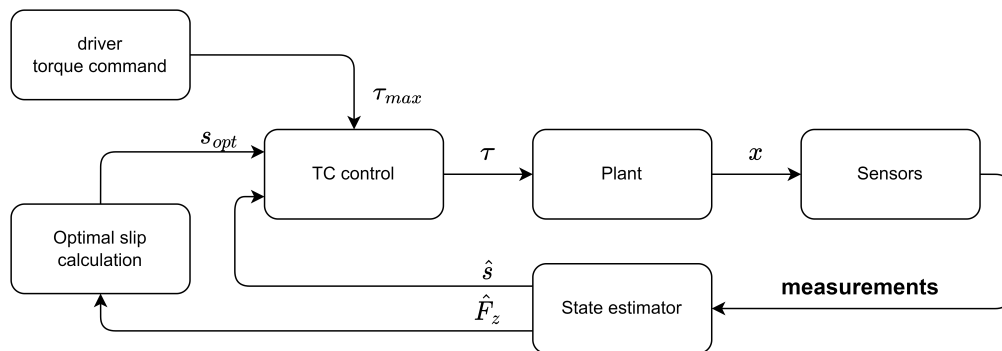


Figure 23: Schema of the Traction control loop

4.2 Optimal slip calculation

In this section, the computation of the optimal slip ratio is discussed. The objective is to maximize the 2-norm of the horizontal friction coefficient, which in turn maximizes the horizontal forces. To visualize this, a traction ellipse (Figure 24) is constructed based on a given normal force, F_z , and slip angle, α . The traction ellipse represents the relationship between longitudinal and lateral friction coefficients as a function of the slip ratio, s .

To construct the traction ellipse, as recommended per an article written by Raymond and Matthew Brach [12], the Nicolas-Comstock equations for horizontal forces correlation, which were presented in

a previous section (Section 2.7.2), are utilized. These equations provide a mathematical model for the correlation between the longitudinal and lateral friction coefficients.

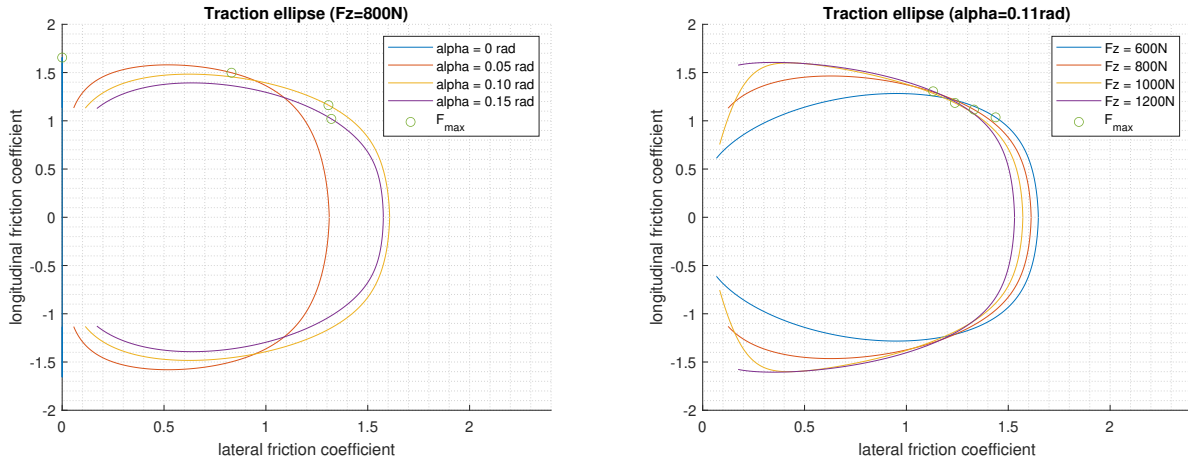


Figure 24: Traction ellipses with a fixed normal force F_z and then a fixed slip angle α

For each traction ellipse, the optimal slip ratio that maximizes the 2-norm of the friction is computed. This optimal slip ratio represents the point on the ellipse that yields the highest possible friction forces.

In order to compute the optimized slip ratio, a 3D map is generated (Figure 25a) that relates the optimal slip ratio to the normal force and slip angle. However, due to uncertainties in the estimation of the vehicle's lateral dynamics, the slip angle is considered to be unknown. To account for this uncertainty, the worst-case scenario is considered, which is represented by a red line on the 3D map.

To visualize this worst-case scenario, a 2D plot is generated by projecting the red line onto a 2D plane (Figure 25b). This plot provides a clear representation of the optimal slip ratio for different normal forces, considering the worst-case slip angle.

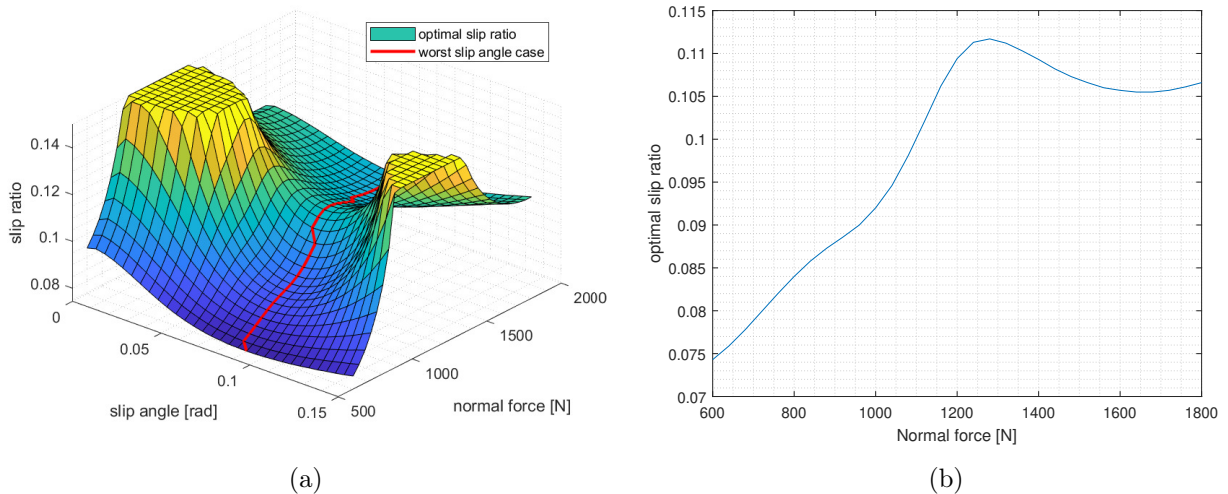


Figure 25: Traction ellipse with a fixed normal force and then a fixed slip angle

Based on the analysis and the 3D map of the optimized slip ratios, it is observed that the optimal slip ratio remains relatively constant around 0.1 for a range of normal forces and slip angles. However, due to the lack of a reliable slip angle estimate, it is decided to choose a fixed slip reference of 0.1. This means that the traction control system will aim to maintain a slip ratio of 0.1 for the wheels in order to maximize the horizontal friction forces and optimize performance.

4.3 Controller

4.3.1 Controller choice

To summarize the previous sections, an estimation of the slip ratio (Section 3) and a slip reference (Section 4.2) were computed. To track this reference, various options were considered. While an MPC controller would be the most efficient due to its ability to handle nonlinear dynamics and constraints, it poses computational challenges for real-time implementation on an embedded computer within a 10ms time frame.

Considering the computational limitations, an explicit NMPC approach was explored, but the large number of variables in the model prevented the construction of a feasible look-up table. Therefore, the idea of a predictive controller was abandoned. Instead, the focus shifted to conceptualizing an optimal PID controller based on simulations.

However, a challenge with the integral gain of a PID controller is that the output rarely corresponds to the applied torque, except for significant accelerations. This leads to the integral gain growing indefinitely when the slip reference is not tracked accurately.

Considering these factors, the best option is to implement a P controller with a feed-forward gain. This approach provides a simpler control structure while still achieving satisfactory performance in tracking the slip reference.

4.3.2 Implementation

In the chosen controller, a control schema with two loops is implemented. The first loop is responsible for computing the maximum torque, while the second loop handles the minimum torque. This approach allows for independent control of the torques applied to each wheel.

The schema of the controller is illustrated below:

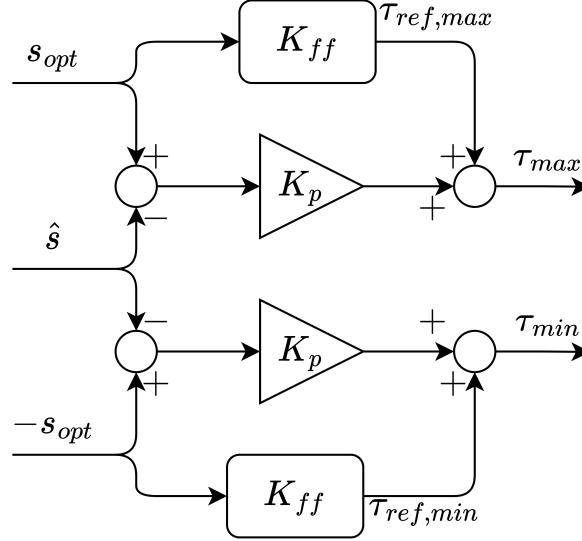


Figure 26: Schema of the Traction control loop

feed-forward : The feed-forward gain used in the controller is determined based on the wheel dynamic equation 7. By solving the equation with the slip reference s_{ref} , the torque can be computed as follows.

$$\tau_{ref} = I_w \dot{\omega} + \tau_f(\omega, P_b) + R_w F_z \mu_x(s_{ref}). \quad (41)$$

Please note that during acceleration, the brake pressure P_b should be set to zero since it is not required. However, during deceleration processes, the brake pressure becomes necessary, especially when computing the minimum torque τ_{min} . The brake pressure is utilized to control the braking force and assist in deceleration.

K_p gain : To track the desired slip ratio, a proportional gain is introduced in addition to the feed-forward gain in the controller scheme. The proportional gain adjusts the control output based on the difference between the desired slip ratio (s_{ref}) and the estimated slip ratio ($\hat{s}$). The control output can be calculated using the following equation:

$$\tau_{min,max} = \tau_{ref} + K_p(s_{ref} - \hat{s}). \quad (42)$$

Certainly, the optimal value for the proportional gain will be discussed in the Testing section (Section 4.4). This is an important parameter that needs to be carefully determined through experimentation and tuning. The appropriate value of the proportional gain depends on various factors such as the vehicle dynamics, tire characteristics, and desired performance of the traction control system.

Torque allocation (without torque vectoring) : To implement the simulation of the traction control, a torque allocation strategy can be used. In this case, since there is no torque vectoring yet, a simplified torque allocation is established. The focus on an optimal allocation based on different controller will be discussed in the next section (Section 5).

The proposed torque allocation strategy is to distribute the torque command to each wheel based on a weighted proportion, taking into account the maximum torque limit obtained from the traction control algorithm. The goal is to ensure that the allocated torque to each wheel is maximized within the limits determined by the traction control system. The equation of the torque allocation is presented below.

$$\tau_i = \min \left(\tau_{max,i}, \frac{\tau_{max,i}}{\sum \tau_{max}} \cdot \tau_{APPS} \right). \quad (43)$$

4.4 Testing

To evaluate the performance of the traction control system, it is recommended to implement it in the actual vehicle and conduct tests during acceleration maneuvers. Initially, tests can be conducted in a straight-line direction to assess the traction control's effectiveness in optimizing slip ratios and maximizing available torque.

Once the straight-line acceleration tests have been completed successfully, additional tests can be performed with constant steering angles to evaluate the system's performance under different driving conditions, such as cornering or maneuvering.

However, since the new LRT4 car is not yet available, simulations can be conducted using the Simulink model developed in the provided appendix. Simulations allow for the evaluation and tuning of the traction control system in a virtual environment, providing insights into its behavior and performance.

The input pedal command used in the simulations is a periodic square signal. As if the pilot applied maximum acceleration and then release the pedal many times.

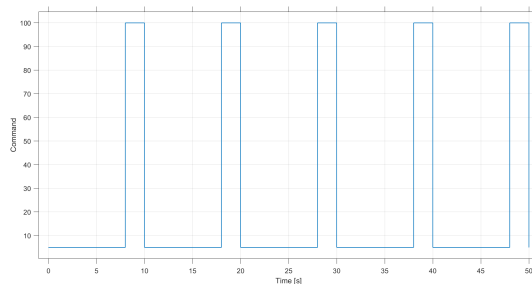


Figure 27: Pedal input for Traction control simulations

Initially, the vehicle is simulated without any traction control, allowing the wheels to freely spin without any intervention. The behavior and performance of the vehicle under the periodic square signal input are observed and analyzed. Next, the traction control system is activated, and different proportional gains are tested to determine their impact on the system's performance. Those results are presented in the following figure.

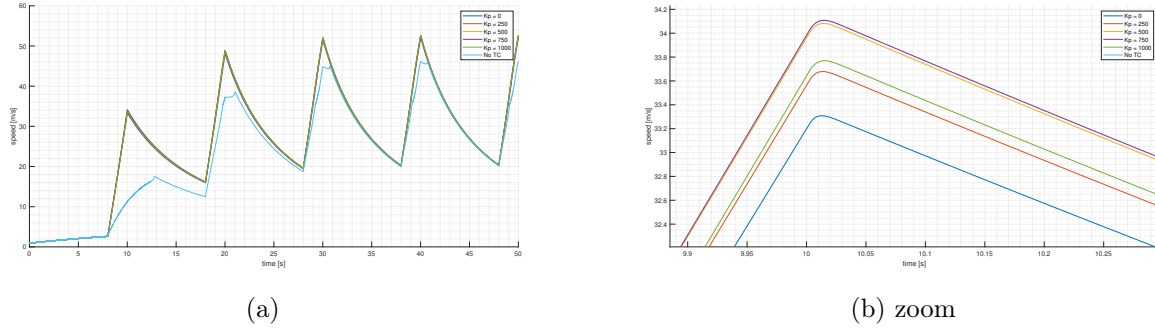


Figure 28: longitudinal velocity in function of the control strategy

The results clearly demonstrate the impact of the traction control system on improving the vehicle's performance. In the absence of traction control, the vehicle experiences over slip, resulting in a lower than desired increase in velocity. This indicates a loss of traction and compromised acceleration.

However, when the traction control system is activated, the situation improves significantly. The controller takes corrective action based on the slip ratio error, and as a result, the vehicle's velocity aligns more closely with the desired value. The traction control system effectively prevents excessive slip, allowing for better traction and improved acceleration.

Finally the difference between the proportional gain effects is quite low but seems to be maximised for a value of $K_p = 750$.

4.5 Further improvements

First, estimating the slip angles can provide valuable information for improving the accuracy of the optimal slip ratio calculation. Slip angles play a crucial role in tire-road interaction and affect the vehicle's handling and traction. By accurately estimating the slip angles, the traction control system can make more informed decisions and adjust the slip reference accordingly, leading to better overall performance.

Additionally, adapting the control gains with respect to velocity and steering inputs can enhance the responsiveness and effectiveness of the traction control system. Different driving conditions and maneuvers may require different control strategies. For example, during high-speed maneuvers or aggressive steering, the optimal slip ratio may vary, and the controller gains can be adjusted accordingly to optimize traction and stability.

5 Control fusion

This section addresses the control fusion problem in the context of a 4WD car. The objective is to determine the allocated torques for each wheel based on inputs from both the traction control and the torque vectoring system, while also considering a power limiter. To effectively combine these control commands, a control fusion algorithm is proposed.

The outputs of the traction control system are the minimum and maximum torques for each wheel, indicating the range within which the torque should be controlled. On the other hand, the torque vectoring system provides an output in the form of the difference between the left and right wheel torques.

To arrive at the final torque command, a fusion algorithm is employed, which takes into account the individual torques from the traction control system, the torque vectoring system, and the power limitation constraints. The fusion algorithm combines these inputs in a manner that ensures the total torque command does not exceed the predetermined power limit, while still considering the desired torque distribution for traction and maneuverability.

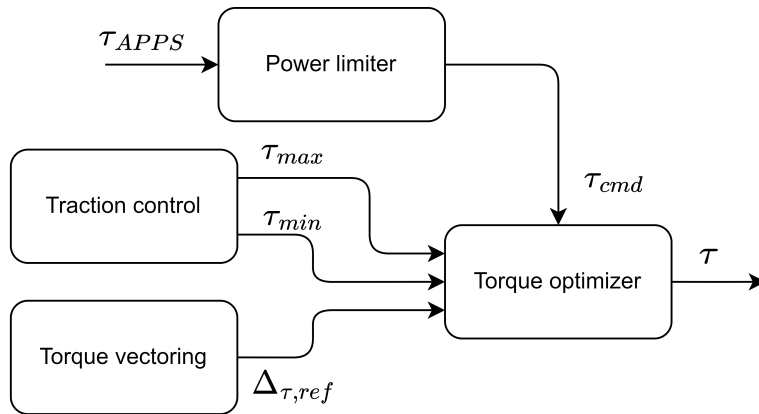


Figure 29: Control fusion schema

5.1 Implementation

5.1.1 Problem formulation

To address the control fusion problem, an optimization problem is formulated. This optimization aims to minimize several objectives. Firstly, it seeks to minimize the difference between the torque command τ_{cmd} and the sum of individual torques assigned to each wheel. This ensures that the desired total torque is closely followed.

Secondly, the optimization aims to minimize the difference between the lateral torque difference $\Delta\tau$ and the torque vectoring command reference $\Delta\tau_{ref}$. This ensures that the desired torque distribution between the left and right wheels, as dictated by the torque vectoring system, is achieved.

Lastly, to prevent situations where excessive torque is applied to either the front or rear wheels, a weighted torque distribution is enforced. This distribution is based on the maximum allowable torques provided by the traction control policy. By incorporating these factors, the optimization ensures that

the torque allocation maintains a balanced distribution among the wheels.

Furthermore, to ensure that the optimization solution adheres to the limits set by the traction control policy, the traction control torque limits τ_{min} and τ_{max} , as well as the torque command τ_{cmd} , are included as hard constraints in the optimization problem. By incorporating these constraints, the optimization algorithm guarantees that the allocated torques for each wheel do not exceed the specified limits and remain within the permissible range.

$$\begin{aligned} \min \quad & w_1(\Sigma_\tau - \tau_{cmd})^2 + w_2(\Delta_\tau - \Delta_{\tau,ref})^2 + w_3 \sum_{i=1}^3 \sum_{j>i} (\Delta_\tau^{ij} - \Delta_{\tau,ref}^{ij})^2 \\ \text{s.t.} \quad & \tau_{min,i} \leq \tau_i \leq \tau_{max,i} \\ & \Sigma_\tau \leq \tau_{cmd} \end{aligned} \quad (44)$$

with

$$\begin{cases} \Sigma_\tau = \tau_{FL} + \tau_{FR} + \tau_{RL} + \tau_{RR} \\ \Delta_\tau = \tau_{FL} - \tau_{FR} + \tau_{RL} - \tau_{RR} \\ \Delta_\tau^{ij} = \frac{\tau_i}{\tau_{max,i}} - \frac{\tau_j}{\tau_{max,j}} \end{cases} \quad (45)$$

To establish the convexity of the control fusion problem formulation, the objective function and constraints are expressed in quadratic form. The quadratic form of the objective function and constraints is a key criterion for determining convexity. The problem formulation can be written as follows:

$$\begin{aligned} \min \quad & (\mathbf{C}\tau - \mathbf{Ref})^T \mathbf{Q} (\mathbf{C}\tau - \mathbf{Ref}) = \tau^T (\mathbf{C}^T \mathbf{Q} \mathbf{C}) \tau - 2\mathbf{Ref}^T \mathbf{Q} \mathbf{C} \tau + \mathbf{Ref}^T \mathbf{Q} \mathbf{Ref} \\ \text{s.t.} \quad & \mathbf{A}\tau \leq \mathbf{b} \end{aligned} \quad (46)$$

Where,

$$\mathbf{A} = \begin{bmatrix} \mathbf{I}_4 & & & \\ & -\mathbf{I}_4 & & \\ 1 & 1 & 1 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} \tau_{max} \\ -\tau_{min} \\ \tau_{cmd} \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 0 & -\frac{\tau_{max,FL}}{\tau_{max,RL}} & 0 \\ 0 & 1 & 0 & -\frac{\tau_{max,FR}}{\tau_{max,RR}} \\ 1 & -\frac{\tau_{max,FL}}{\tau_{max,FR}} & 0 & 0 \\ 0 & 0 & 1 & -\frac{\tau_{max,RL}}{\tau_{max,RR}} \end{bmatrix}. \quad (47)$$

And,

$$\mathbf{Ref} = \begin{bmatrix} \tau_{cmd} \\ \Delta_{\tau,ref} \\ \mathbf{0}_{4 \times 1} \end{bmatrix}. \quad (48)$$

It is then straight forward to prove the problem convexity since the matrix $\mathbf{C}$ is full rank.

The matrix $\mathbf{Q}$ can be adjusted, and a choice needs to be made regarding the priority given to torque vectoring compared to other optimization variables, as it should be consistently honored. Therefore, a potential $\mathbf{Q}$ matrix is provided below.

$$\mathbf{Q} = \text{diag}([1 \cdot 10^{-3} \quad 1 \quad 1 \cdot 10^{-2} \quad 1 \cdot 10^{-2} \quad 1 \cdot 10^{-2} \quad 1 \cdot 10^{-2}]). \quad (49)$$

5.1.2 Computational time

Reducing computational time is the primary concern when addressing an optimization problem, as it must be completed within 10ms to be suitable for implementation on an embedded computer. To address this challenge, several proposals are being discussed in this paragraph with the aim of minimizing the computational time.

Starting point : One approach to reduce computational time is to find a starting point that is close to the optimal solution. It is generally faster to converge to the optimal point from a starting point that is closer to it rather than starting from $\mathbf{0}$. One idea is to begin from the optimal value obtained in the previous optimization problem. However, since the constraints can vary between problems, there is no guarantee that this starting point will fall within the constraints of the current problem. For instance, if the driver releases the acceleration pedal, the torque command τ_{cmd} becomes zero, whereas it may not have been the case in the previous optimization problem.

Another proposition is to start with an equal distribution of the torque command on each wheel and maximize it using the maximum allowable torques determined by the traction control. This approach ensures that the starting point remains within the constraints while attempting to optimize the torque command distribution. By starting from a feasible point, the computational time can be reduced, and the optimization process can proceed more efficiently. The formulation of the proposed starting point is describe in equation (50).

$$\tau_{0,i} = \min(\tau_{APPS}/4, \tau_{max,i}). \quad (50)$$

STOP condition : Another strategy to decrease computational time involves implementing a stopping condition within the optimization algorithm. It is not necessary to determine the applied torque with a precision of 0.1Nm. Therefore, a condition can be added where if the evolution of the minimization function falls below a certain threshold ϵ , the current values are retained without further optimization.

By introducing this stopping condition, the algorithm can terminate the optimization process once the desired level of optimization has been achieved, eliminating the need for additional iterations that may not significantly improve the results. This helps to reduce the computational time required while still ensuring that the optimization meets the required level of precision. The equation of the stopping condition can be expressed as following.

$$\text{STOP if } |J^*(k) - J^*(k-1)| < \epsilon. \quad (51)$$

5.2 Testing

In this particular test, only the evolution of computational time is being evaluated. The implementation of the new 4WD car has not yet been tested. It is important to note that the computational time tests are conducted on a personal computer operating in a low efficiency mode. Therefore, the results obtained from these tests cannot be directly compared to the computational time expected on the embedded computer.

The purpose of these tests is to demonstrate the variation in computational time based on the parameters discussed earlier. The focus is on understanding how different factors impact the time required for computation rather than directly simulating the performance on the embedded computer. These tests serve to highlight the differences in computational time and provide insights into optimizing the algorithm for faster execution when implemented on the embedded computer.

This test involves calculating the average time for 1000 optimization problems. A fixed stop condition is set at $\epsilon = 0.01$, which has been determined to be a suitable value in terms of maintaining an acceptable level of accuracy. The purpose of conducting 1000 optimization problems and measuring the average time is to obtain a robust estimation of the computational performance.

	original	τ_0 only	STOP only	τ_0 and STOP
time [ms]	8.05	8.94	6.44	4.26

Table 4: Comparison of optimization computational time

It is evident that activating both propositions, namely finding a starting point close to the optimal point and implementing a stop condition, can lead to a significant reduction in computational time. This reduction by a factor of 2 is noteworthy, especially when considering the implementation on an embedded computer.

5.3 Further improvements

To achieve further improvements, two additional measures can be considered. Firstly, conducting a test of the formulation efficiency either through simulation or on the actual car can provide valuable insights. This test would evaluate the efficiency of the formulation itself, identifying potential bottlenecks or areas for optimization.

Secondly, conducting deeper research on finding an optimal starting point can lead to additional reductions in computational time. By investigating and developing techniques for identifying starting points that are closer to the optimal solution, the convergence time of the optimization algorithm can be significantly improved.

6 Conclusion

In conclusion, this semester project focused on extensive research into vehicle dynamics. The study revealed that vehicles exhibit numerous nonlinearities arising from factors such as tire characteristics and braking scenarios. Fortunately, the presence of multiple sensors on the vehicle enabled the validation of the proposed model. The sensors also highlighted the significance of wheel frictions in the overall dynamics.

The second part of the project delved into the state estimation of the vehicle. Leveraging the information provided by the available sensors and incorporating knowledge of the vehicle dynamics proved beneficial. While certain states, such as normal forces (F_z), were relatively easy to estimate, slip ratios and car velocity posed more challenges. Given the absence of an accurate velocity sensor with a high refresh rate (100Hz), a mixed Kalman filter (MKF) was implemented to effectively utilize the sensor data, the model, and their respective accuracies. Accurate slip estimation is crucial for effective traction control, as without it, only open loop control is feasible.

The control strategy was discussed, leading to the conclusion of adopting a proportional controller with a feed-forward gain. Additionally, a control fusion algorithm was explored to address traction control, torque vectoring, and power limitation. While an optimization problem was formulated, it has yet to be tested. However, a thorough discussion on computational time optimization was conducted, exploring ways to enhance its speed.

Overall, this project has significantly contributed to understanding vehicle dynamics, state estimation, control strategies, and computational time optimization.

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A Parameters

A.1 Physical parameters

Chassis

parameter	LRT3	LRT4	unit
a	1.24	1.24	m
b	1.20	1.24	m
l_f	0.780	0.785	m
l_r	0.795	0.785	m
l	1.575	1.570	m
m_{car}	223	230	kg
m_{tot}	303	300	kg
z_{cg}	0.288	0.295	m
I_{zz}	180	?	Nms ²

Aerodynamics

parameter	LRT3	LRT4	unit
A_f	1.06	1.61	m ²
c_{drag}	1.28	0.97	
c_{lift}	3.12	2.33	
ρ_{air}	1.2	?	kg/m ³

Motor

parameter	LRT3	LRT4	unit
τ_{max}	90	21	Nm
η	4.5	13.188	

Wheel

parameter	LRT3	LRT4	unit
R_w	0.202	0.202	m
I_w	0.3	0.3	Nms ²
k_d	0.17	?	s
k_s	15	?	

Tire

μ_{long}		μ_{lat}	
b_1	$-6.75 \cdot 10^{-6}$	b_1	$3.79 \cdot 10^1$
b_2	$1.35 \cdot 10^{-1}$	b_2	$5.28 \cdot 10^2$
b_3	$1.20 \cdot 10^{-3}$		
c_1	1.86	c_1	1.57
d_1	$1.12 \cdot 10^{-4}$	d_1	$-2.03 \cdot 10^{-4}$
d_2	1.57	d_2	1.77
e_1	$-5.38 \cdot 10^{-6}$	e_1	$-2.24 \cdot 10^{-3}$
e_2	$1.11 \cdot 10^{-2}$	e_2	1.81
e_3	-4.62		

parameter	LRT3	LRT4	unit
BCD_s	36.84	36.84	
B_s	11.15	11.15	
C_s	1.98	1.98	
D_s	1.67	1.67	
E_s	0.97	0.97	

A.2 Estimation parameters

parameter	LRT3	LRT4	unit
$\sigma_{\eta_{a_x}}$	-	0.02	$\text{m/s}^2/\text{Hz}^{1/2}$
$\sigma_{\eta_{a_y}}$	-	0.2	$\text{m/s}^2/\text{Hz}^{1/2}$
$\sigma_{\eta_{\dot{\psi}}}$	-	$2 \cdot 10^{-5}$	$\text{rad/s}/\text{Hz}^{1/2}$
$\sigma_{\eta_{\omega}}$	-	0.05	$\text{rad/s}/\text{Hz}^{1/2}$
$\sigma_{\eta_{\tau}}$	-	20	$\text{Nm}/\text{Hz}^{1/2}$
$\sigma_{\xi_{\dot{v}_x}}$	-	0.03	$\text{m/s}^2/\text{Hz}^{1/2}$
$\sigma_{\xi_{\dot{v}_y}}$	-	0.1	$\text{m/s}^2/\text{Hz}^{1/2}$
$\sigma_{\xi_{\dot{a}_x}}$	-	0.02	$\text{m/s}^3/\text{Hz}^{1/2}$
$\sigma_{\xi_{\dot{a}_y}}$	-	0.2	$\text{m/s}^3/\text{Hz}^{1/2}$
$\sigma_{\xi_{\ddot{\psi}}}$	-	$1 \cdot 10^{-3}$	$\text{rad/s}^2/\text{Hz}^{1/2}$
$\sigma_{\xi_{\dot{s}_i}}$	-	0.03	$\text{Hz}^{-1/2}$

A.3 Traction control parameters

parameter	LRT3	LRT4	unit
K_p	-	$\simeq 750$	

B Simulink description

To evaluate the algorithms discussed in this report, a Simulink model has been created. The purpose of this model is threefold: firstly, to simulate the car with non-linear characteristics; secondly, to assess the state estimator in a post-processing manner; and thirdly, to test a closed-loop traction controller. This appendix provides instructions on utilizing Simulink and examines the conducted tests.

B.1 Common

The Simulink model in question is organized using projects in Matlab, which enable the creation of multiple connections between files. The model consists of several sub-projects, the first being called **Common**. This sub-project contains all the necessary data for testing, such as vehicle parameters and various processing functions.

To utilize the data, a ".mat" table must be generated using the `csvParser.m` file. The vehicle parameters are organized in dictionaries and stored in folders named after their respective car names. The simulation requires the relevant parameters to be written in the **VehParams** folder. Parameters used in simulation should be written in the **VehParams** folder.

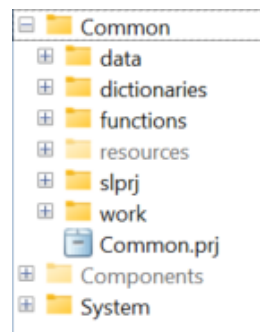


Figure 30: common file classification

B.2 Components

The folders are further divided into components, which include the vehicle model and the state estimator, among others. Additionally, there is a system folder that encompasses the closed-loop system, comprising all the components. Each of these folders follows the same classification shown in Figure 31. There is a **models** folder, which houses all the Simulink sub-models, a **functions** folder primarily containing initialization functions, a **dictionaries** folder that stores all the relevant data for a specific model, and finally, a **tests** folder that contains test files managed by Matlab's test manager.

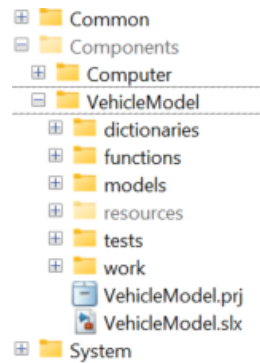


Figure 31: VehicleModel files classification

B.2.1 Vehicle Model

The vehicle model receives three inputs: the steering angle, the motor currents, and the brakes pressure. This model is divided into three main parts. First, preliminary transformations are implemented. Then, the dynamics of each individual wheel are modeled. Finally, a General dynamics block is used to implement the second law of Newton.

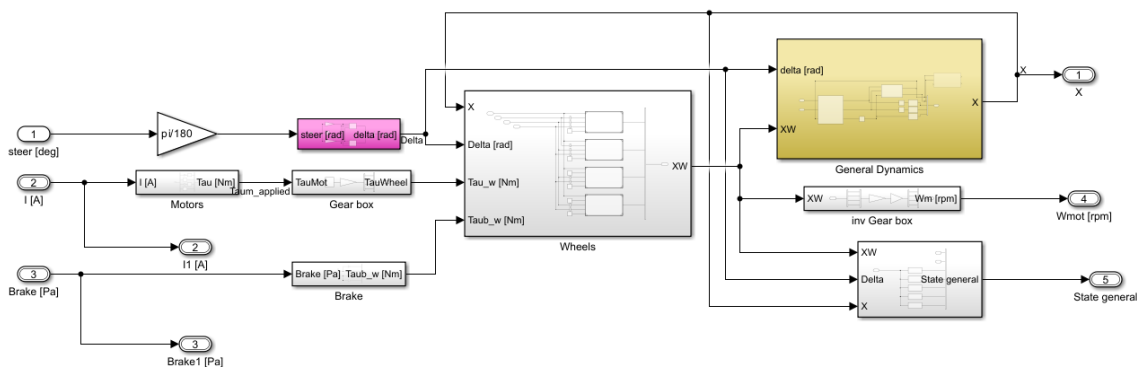


Figure 32: Vehicle Simulink Model

To open the model used for testing with specific inputs, follow these steps:

1. Right-click on the desired location within the Simulink model.
2. Select **Test Harness** from the context menu.
3. Choose **Manage Test Harnesses** from the submenu.
4. Select **ModelValidation** from the list of available test harnesses.

By performing these steps, the model used for testing will be opened, allowing you to provide the required inputs. The corresponding test file associated with this process is named **ModelValidation.mldatx**.

B.2.2 State estimation

The state estimation is included within the **Computer** folder. The model takes the following structure (Figure 33), where the initial step involves performing the preliminary transformations discussed in

this report (Section 3.2). Subsequently, the cyan sub-model is responsible for executing the state estimation.

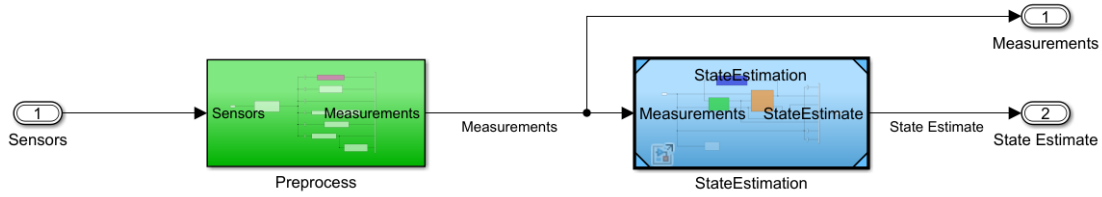


Figure 33: State estimation Simulink Model

When clicking on the State estimation sub-model, a subsequent model is displayed (Figure 34). The green components represent the overall state estimation, which is based on measurements and previous estimates, as described in Section 3.3. Following that, the orange component implements the Kalman Filter. The associated test file is named `StateEstimation.mldatx`.

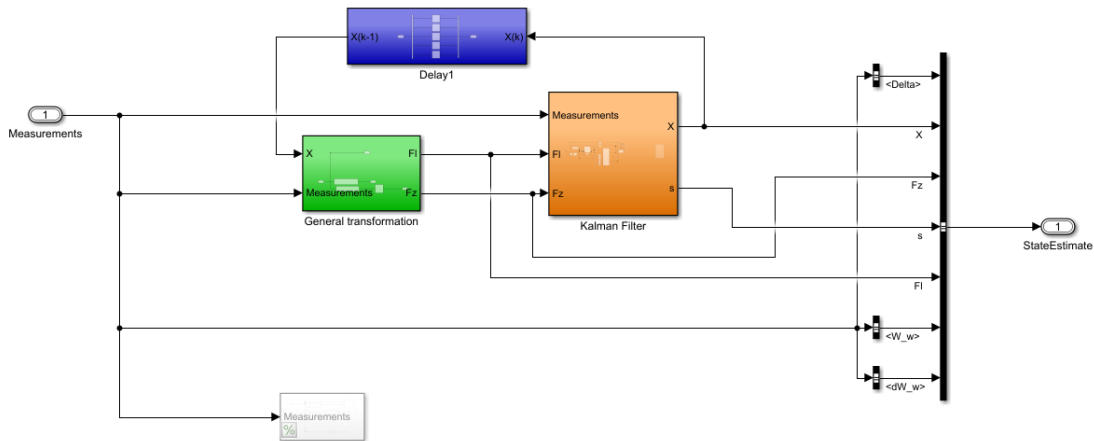


Figure 34: State estimation Simulink Model

B.2.3 Control

In the same folder, the control part is available. The corresponding model (Figure 35b) includes the Torque vectoring component, indicated in cyan, as discussed in Nathan Decurnex's report [2]. The blue subsystem implements the traction controller, as explained in Section 4. Additionally, the red subsystem incorporates the Control fusion described in Section 5.

To facilitate testing, a small application (Figure 35a) has been developed to modify the control mode of the simulation. This application also allows for tuning the K_p gain of the traction control and the K_u gain (understeer gain) of the torque vectoring. To open this application, please execute the command `ControlBoard` in the Matlab terminal.

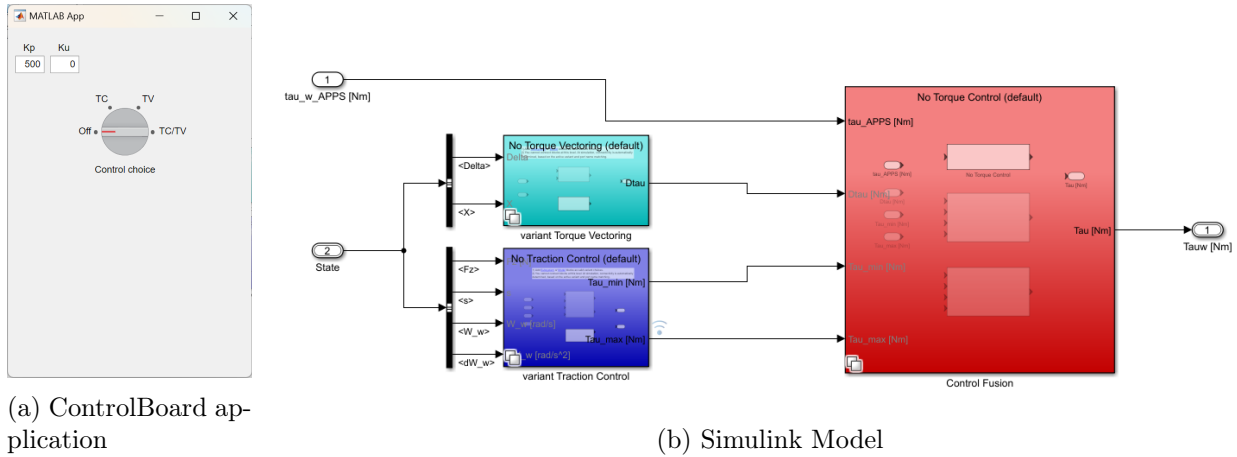


Figure 35: Control Simulink Model

B.3 Close Loop

To evaluate the traction control strategy through simulation, the closed loop is implemented in the **System** folder. This model incorporates the previous vehicle model, sensors, and calculations involving estimation and control algorithms. Discrete signals are represented by blue lines, while continuous signals are depicted in black. The corresponding test file for this system is named `TC_test.mldatx`.

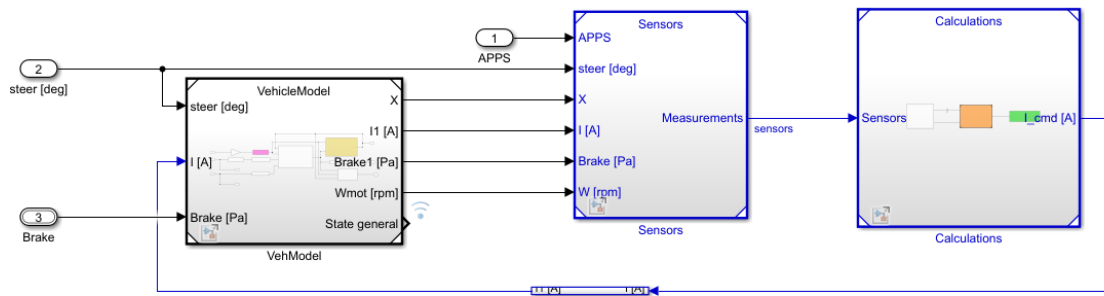


Figure 36: Vehicle Model Simulink

C Labview description

The embedded computer is responsible for implementing all the algorithms related to state estimation and control. The code is written in Labview, and this section aims to provide an explanation of how the aforementioned algorithms are put into practice.

C.1 State estimation

The state estimation can be found in a unique subVI which looks like the following :



Figure 37: State estimation SubVI

The implementation utilizes two global variable files: `Estimation_variables` and `VehParams`. The former contains the variables to be estimated, while the latter holds the vehicle parameters. The state estimation process is divided into three parts, as outlined in Section 3. The first part involves preliminary transformations (refer to Figure 40), the second part comprises general estimations (refer to Figure 38), and the final part involves the use of the Kalman Filter (refer to Figure 39).

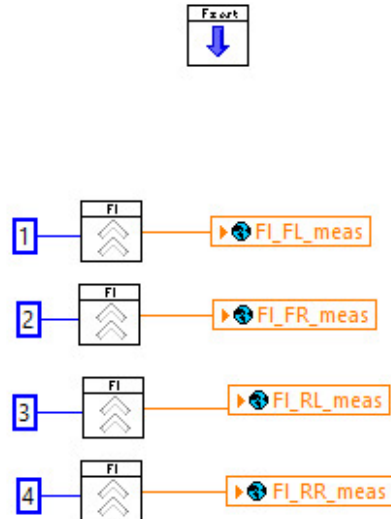


Figure 38: General estimations implementation



Figure 39: Kalman filter implementation

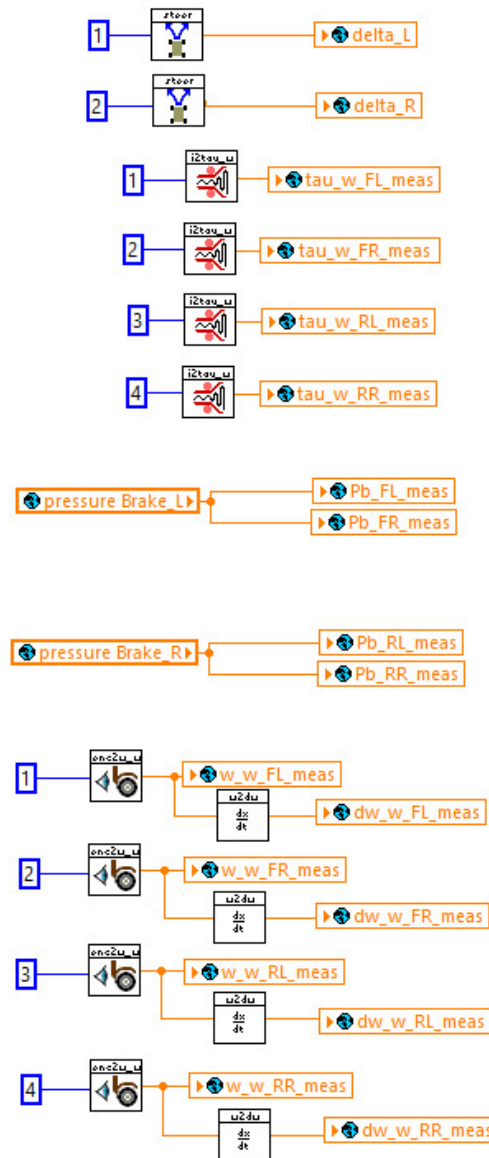


Figure 40: Preliminary transformations implementation

C.2 Traction Control

The implementation of traction control is carried out within the `magic_cycle` subVI. A separate controller is designed for each wheel, as depicted in Figure 41. To calculate the maximum and minimum allocated torques, a proportional controller with feedforward is employed, as illustrated in Figure 42.

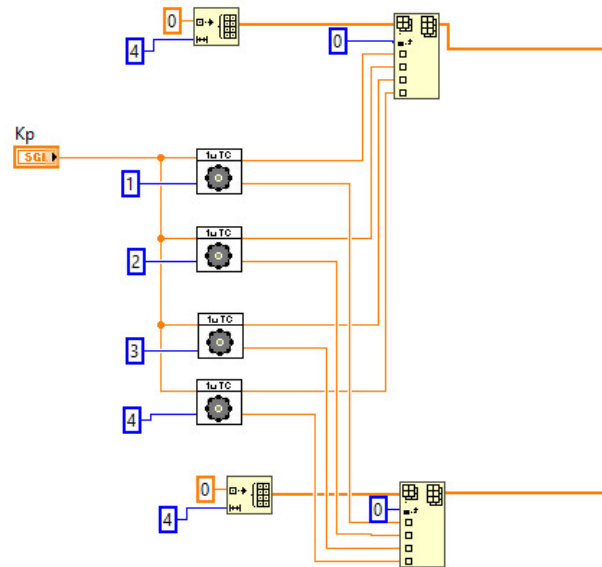


Figure 41: Traction control implementation

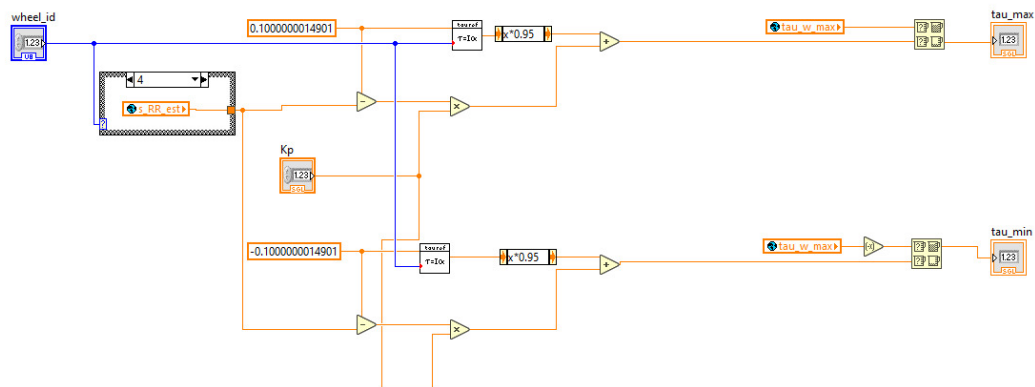


Figure 42: Proportional controller implementation

D Kalman Filter without slip ratios

$$\mathbf{x} = [v_x \ v_y \ a_x \ a_y \ \dot{\psi}]^T. \quad (52)$$

$$\dot{\mathbf{x}} = \begin{cases} \dot{v}_x = a_x + v_y \dot{\psi} + \xi_{\dot{v}_x} \\ \dot{v}_y = a_y - v_x \dot{\psi} + \xi_{\dot{v}_y} \\ \dot{a}_x = \xi_{\dot{a}_x} \\ \dot{a}_y = \xi_{\dot{a}_y} \\ \dot{\psi} = \xi_{\dot{\psi}} \end{cases} \Rightarrow \hat{\mathbf{x}} = \begin{cases} \hat{v}_x = \hat{a}_x + \hat{v}_y \hat{\psi} \\ \hat{v}_y = \hat{a}_y - \hat{v}_x \hat{\psi} \\ \hat{a}_x = 0 \\ \hat{a}_y = 0 \\ \hat{\psi} = 0 \end{cases}. \quad (53)$$

$$\mathbf{A}_{\text{ekf}} = \begin{bmatrix} 0 & \hat{\psi} & 1 & 0 & \hat{v}_y \\ -\hat{\psi} & 0 & 0 & 1 & -\hat{v}_x \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{B}_{\text{ekf}} = \mathbf{I}_5, \quad \mathbf{Q}_{\text{ekf}} = \text{diag}([\sigma_{\xi_{\dot{v}_x}}^2, \sigma_{\xi_{\dot{v}_y}}^2, \sigma_{\xi_{\dot{a}_x}}^2, \sigma_{\xi_{\dot{a}_y}}^2, \sigma_{\xi_{\dot{\psi}}}^2]). \quad (54)$$

$$\mathbf{C}_{\text{lkf}} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{D}_{\text{lkf}} = \mathbf{I}_3, \quad \mathbf{N}_{\text{lkf}} = \text{diag}([\sigma_{\eta_{a_x}}^2, \sigma_{\eta_{a_y}}^2, \sigma_{\eta_{\dot{\psi}}}^2]). \quad (55)$$

In the considered problem, wheels speed measurements are compared to the model. But the wheel speed estimate needs the total torque measurements to work. Hence a solution to consider it is to augment the state by adding the total torque measure. The new augmented state is the following.

$$\hat{\mathbf{x}} = \begin{bmatrix} \hat{\mathbf{x}} \\ \tilde{\tau}_{t,i} \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \Sigma & 0 \\ 0 & \sigma_{\tau_{tw}}^2 \end{bmatrix}. \quad (56)$$

Using this state, measure and estimate can be found using the non linear dynamics of the car. Here is a summary of the used equations.

$$\begin{cases} \tilde{y} = \tilde{\omega}_{w,i} \\ \hat{y} = h(\hat{\mathbf{x}}) = \frac{\hat{s}_i + 1}{R_w} \hat{v}_{i,long} \end{cases}. \quad (57)$$

Where the longitudinal velocity of the wheel is found from the state.

$$\hat{v}_{i,long} = (\cos \delta_i \ \sin \delta_i) \cdot \begin{pmatrix} \hat{v}_x - p_y \hat{\psi} \\ \hat{v}_y + p_x \hat{\psi} \end{pmatrix}. \quad (58)$$

Finally, in order to reduce the computational cost. the inversion of the magic formula is done only in the linear domain ($s < 0.01$). K_s is the stiffness and found as the linear coefficient of the magic formula.

$$\hat{s}_i = \frac{\tilde{\tau}_i}{R_w K_s \hat{F}_{z,i}} \text{ with } K_s = BCD_x = (b_1 \hat{F}_{z,i}^2 + b_2 \hat{F}_{z,i}) e^{-b_3 \hat{F}_{z,i}}. \quad (59)$$

Despite this linearization, a way to keep the measure precision is to only apply this UKF if the estimate slip is lower than 0.01 and only take into account the wheel with the less slip ratio in order to the linear model to be write. It is important to precise that the slips ratios are 50% of the time below than 0.01.